



<Customer Name> Oracle Database@AWS

Solution Definition

28th of May 2026 | Version 1.0

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Guide:

Document Control

Guide:

1.1 Version Control

Guide:

Example:

Version	Authors	Date	Comments
0.1	DB@AWS Team	1st October 2025	Published first version of SDD.
0.2	DB@AWS Team	January 6th, 2026	Revised version of SDD.

1.2 Team

Guide:

Example:

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1.3 Document Purpose

Guide:

Example:

Business Context

Guide:

Example

MYTestCustomer is a large airline operator, looking to modernize their application platform. Through this modernization, MYTestCustomer aims to improve the customer experience, enhance performance by 30%, cope with the larger forecasted transaction volume, and simplify future expansion into additional geographies.

2.1 Executive Summary

Guide:

A section describing the Oracle differentiator and key values of the solution of our solution for the customer, allowing the customer to make decisions quickly.

Example

Over the past year, MYTestCustomer has embarked on a strategic transformation initiative to migrate its IT assets to the public cloud, aligning the broader goals of agility, scalability and operational efficiency. As part of this initiative, AWS cloud has been selected as the primary public cloud provider for MYTestCustomer.

Key workloads within this transformation are the ERP and Business Intelligence (BI) ecosystem, which are currently hosted on an on-premises Oracle Exadata X7 infrastructure.

This environment supports two mission critical Oracle 19c Container Databases (CDBs)

- A highly transactional OLTP database that underpins the ERP system
- A data warehouse (DWH) database supporting a Business Intelligence solution. The database is optimised for analytics through features such as partitioning, parallel query execution, and Hybrid Columnar Compression (HCC)

To ensure continuity and performance parity in the cloud, MYTestCustomer has selected Oracle Database@AWS as the foundation for migrating these databases. This solution enables MYTestCustomer to run Oracle database on Exadata infrastructure within an AWS data center, delivering Exadata-level performance and advanced Oracle capabilities without the burden of hardware ownership or infrastructure management.

Oracle Database@AWS offers a seamless Multicloud approach that preserves enterprise-grade database performance while advancing MYTestCustomer's Multicloud strategy.

This Multicloud deployment provides the following benefits:

- The elasticity and innovation of Oracle Database@AWS
- Robust performance and advanced features of Oracle Exadata
- Simplified operations through managed services
- Reduced infrastructure overhead and TCO

2.2 Workload Business Value

Guide:

A clear statement of specific business value as part of the full workload scope. Try to keep it SMART: Specific, Measurable, Assignable, Realistic, and Time-Related - Agree on the business value with the customer. Keep it business-focused, and speak the language of the LoB which benefits from this Workload: "Increase Customer Retention by 3% in the next year" or "Grow Customer Base with Executive Decision-Making from our Sales and Support Data". Avoid technical success criteria such as "Migrate App X to Oracle Cloud" or "Provision 5 Compute Instances". Avoid Oracle success criteria and language "Get Workload Consuming on OCI".

Example

The workloads in the scope are of vital importance for MYTestCustomer. Both the OLTP workload and the DWH workloads are needed for the company operation and any disruption in the applications would be costly and would badly affect customer satisfaction. Furthermore, a recent security assessment and a review of the Disaster Recovery plan has revealed that the current architecture must be improved in terms of security and resilience. For this reason, MYTestCustomer has decided to move these workloads to AWS, and leverage the Oracle Database@AWS Exadata solution. The main reasons for this migration to Oracle Database@AWS are:

- Leverage the availability of Exadata database services in AWS to achieve the best database performance (AWS is preferred choice of MYTestCustomer for application's tier)
- Reduce the TCO leveraging the dynamic scaling capabilities of Database resources, and be able to scale dynamically without buying new hardware
- The possibility to transfer existing Oracle licenses to the cloud (BYOL)

- Avoid the cost of maintaining the underlining Exadata infrastructure and re-focusing the IT personnel on more business-oriented tasks
- Implement a Disaster Recovery strategy, leveraging the availability of Oracle Database@AWS in multiple regions, and multiple availability zones within each region
- Generational performance improvements by going from Exadata X7 to Exadata X11M

Workload Requirements and Architecture

3.1 Overview

Guide:

Describe the Workload: What applications and environments are part of this Workload migration or new implementation project, and what are their names? The implementation will be scoped later and can be a subset of the Solution Definition and proposed overall solution. For example, a Workload could exist of two applications, but the implementer would only include one environment of one application. The workload chapter is about the whole Workload and the implementation scope will be described later in the chapter [Solution Scope](#).

You can describe an existing solution in the optional chapter for a Current State Architecture , which is typically located after the Non functional requirements and before the Future State Architecture.

Example

Currently, the database environment is deployed on a single Exadata system, while the application servers for both the ticketing application and BI application are deployed on a virtual machines within a VMWare estate. The batch processes that update the DWH database run as custom Python programs on number of Linux hosts, also hosted on the same VMWare estate.

Below is the list of databases with their existing known versions, platforms, size, life cycle environment and virtual cluster.

Database Name	Database Version	Database Platform	Database Size	Env.	Cluster
OLTP	19.19	Exadata (Linux x86-64)	1.2 TB	Prod	clu-oltp-prod
DWH	19.19	Exadata (Linux x86-64)	7 TB	Prod	clu-dwh-prod
OLTP-TEST	19.19	Exadata (Linux x86-64)	700 GB	Non Prod	clu-oltp-nonprod
OLTP-DEV	19.19	Exadata (Linux x86-64)	500 GB	Non Prod	clu-oltp-nonprod
DWH-TEST	19.19	Exadata (Linux x86-64)	3 TB	Non Prod	clu-dwh-nonprod
DWH-DEV	19.19	Exadata (Linux x86-64)	1 TB	Non Prod	clu-dwh-nonprod

3.2 Non-Functional Requirements

Guide:

3.2.1 Regulations and Compliances Requirements

Guide:

This section captures specific regulatory or compliance requirements for the solution. These may limit the types of technologies that can be used and may drive some architectural decisions.

*The Oracle Cloud Infrastructure Compliance Documents service lets you view and download compliance documents:
<https://docs.oracle.com/en-us/iaas/Content/ComplianceDocuments/Concepts/compliancedocsoverview.htm>*

However, the Oracle Database@AWS solution provides a part of OCI inside AWS. Therefore, any compliance documents relevant to either AWS or OCI do not automatically apply to Oracle Database@AWS. It is vital that any requirements are captured so that they can be checked against assessments already done, or on the roadmap.

What needs to be captured are any requirements of the Oracle Database@AWS elements, and anything running in OCI. Requirements for specific elements of the solution that will be running completely in AWS (for example VMware Cloud on AWS) do not have to be captured here.

The compliance certifications and service management responsibilities for Oracle Database@AWS [can be found in the Oracle documentation](#)

If there are no requirements at this stage, then please state it. Leave the second sentence as a default in the document.

Example:

At the time of this document creation, no Regulatory and Compliance requirements have been specified.

In addition to these requirements, the latest version of the [CIS Oracle Cloud Infrastructure Foundation Benchmark v1.2](#) will be applied to the Customer OCI tenancy.

3.2.2 Environments Required in AWS

Guide:

A diagram or list detailing all the required environments (e.g. development, test, live, production, etc.). Focus on the Oracle Database@AWS/OCI elements, but also keep in consideration how they interact with the rest of the environments in AWS.

As a minimum we should capture the Exadata hardware requirements of Oracle Database@AWS in this section.

If you like to describe a current state, you can use or add the chapter 'Current State Architecture' before the 'Future State Architecture'.

Example:

Production and DR will each require a separate ¼ rack Exadata in different availability zones within the same AWS region eu-central-1 (linked to OCI region Frankfurt). Dev and Test are co-located on another ¼ rack Exadata.

The Production and DR Exadata infrastructures will host different VM clusters for each of the databases. The Dev/Test databases will share clusters between workload types.

Name	Size	Location	DR	High Availability	RPO	RTO
Prod	Quarter rack X11M	AWS eu-central-1	Yes	Yes	0	2 hrs
Dev + Test	Quarter rack X11M – 50% of Prod size	AWS eu-central-1	No	Yes	24 hrs	24 hrs
DR	Quarter rack X11M – 100% of Prod size	AWS eu-central-1	No	Yes	4 hrs	12 hrs

3.2.3 Oracle Database@AWS High Availability and Disaster Recovery Requirements

Guide:

This section captures the resilience and recovery requirements for the solution. Note that these may be different from the current system.

The Recovery Point Objective (RPO) and Recovery Time Objective (RTO) requirement of each environment should be captured in the environments section above, and wherever possible.

- What are the RTO and RPO requirements of the Application?
- What are the SLAs of the application?

- *What are the backup requirements*

Note that if needed, this section may also include an overview of the proposed backup and disaster recovery proposed architectures. Due to the Multicloud nature of the Oracle Database@AWS solution, there can be considerable traffic between AWS and OCI datacenters in Disaster Recovery scenarios, which can incur data transfer cost. For more information see chapter 'Oracle Database@AWS High Availability and Disaster Recovery' under 'Solution Considerations'.

This chapter is mandatory, while there could be no requirements on HA/DR, please mention that in a short single sentence.

Example:

At the time of this document creation, no Resilience or Recovery requirements have been specified.

3.2.4 Backup and Recovery

Guide:

This section defines the backup and recovery needs for the workload, which may differ from the current system.

RPO/RTO: Specify the Recovery Point Objective (RPO) and Recovery Time Objective (RTO) for each environment.

Backup Retention: Define retention periods based on compliance, business and operational needs.

Database or Storage Server Expansions: If not applicable to storage/database expansion, mark as "N/A due to limited-scope activity: Only Expansion Needed"

Example:

While the current backup solution uses a combination of backup technologies, such as NFS, the target solution will be upgraded with Zero Data Loss Recovery Appliance to accommodate increasing demands for Ransomware Protection (Compliance and Immutability) and Zero RPO.

Environment	Backup Frequency	Retention	Recovery Time	Recovery Point	Backup Target	Retention Lock
Production	Daily, Incremental forever	60 days	2 hours	Zero	Recovery Service	Yes
Dev + Test	Daily, Incremental, Weekly Full	30 days	24 hours	30 m	Object Storage	N/A
DR	Daily, Incremental forever	60 days	2 hours	30 m	Recovery Service	Yes

ExaDB-D cloud tooling supports OCI Object Storage, AWS S3 and Autonomous Recovery Service as backup solution. Please note:

- ADB-D does not support Recovery Service as a target, only OCI Object Storage and AWS S3.
- ADB-D does support X-region replication of backups on OCI OSS only.
- ADB-D does support retention lock of backups on OCI OSS And AWS S3.
- LTR is not supported on Recovery Service @AWS, if LTR is required OCI OSS And AWS S3 must be used.

Note that special consideration needs to be given for to the specifics of HA/DR in Oracle Database@AWS. Specifically:

- *The regional availability of the solution: <https://docs.aws.amazon.com/odb/latest/UserGuide/getting-started.html#supported-odb-regions>*
- *Differences in naming conventions (Availability Zones in AWS rather than the OCI Availability Domains)*

- *Note that Autonomous Recovery Service is also available either @OCI or @AWS, the customer has the choice to leverage RCV in either AWS or OCI. Oracle Database Autonomous Recovery Service is the recommended backup service for automated ExaDB-D backup. All backup options will draw down on a customer's AWS commitments.*

If there are any (security) requirement to always keep all data within the AWS data centers, then DR and Backup solutions need to be considered with some care, and this would have to be captured in this section. Recovery Service is the only solution to satisfy that requirement.

This chapter is mandatory, while there could be no requirements on HA/DR, please mention that in a short single sentence.

Example:

The databases will be implemented on 2-node Exadata clusters, thereby providing high availability. Oracle MAA Best Practices and Reference Architectures should be considered for all environments.

Both applications have a RPO of 0, with an RTO of 5 minutes. For this reason, Active Data Guard will be used to keep the Production and DR databases in sync. The traffic between the Primary and Standby databases will run through the AWS-Managed network.

In addition, OCI Autonomous Recovery service (with Real-time data protection, i.e. Zero Data Loss mode) in the AWS data center will be used to accommodate increasing demands for Ransomware Protection (Compliance and Immutability) and near-zero RPO.

3.2.5 Security Requirements

Guide:

Capture the Non-Functional Requirements for security-related topics. The requirements can be (but don't have to be) separated into:

- *Identity and Access Management*
- *Data Security*

Other security topics, such as network security, application security, key management, or others can be added if needed.

Take note that parts of the solution running on AWS but which are not directly related to Oracle Database@AWS, are not required to be captured here. For example, AWS application security does not have to be captured here.

For Oracle Database@AWS there are a number of specific considerations which can be clarified during this phase, or later. For more details on the following requirements you can check this [Oracle documentation section](#)

- OCI access for (Database) Management
- Oracle Database@AWS onboarding requirements (required privileges)
- AWS governance: Capture AWS accounts/organisation design, resource groups and users/groups details. E.g. current application environments accounts to be used with Oracle Database@AWS deployment. Make sure to compare with limitations of Oracle Database@AWS in terms of mapping to accounts.
- Identity and Access Management (IAM)

3.2.5.1 Identity and Access Management

Guide:

You can capture all the Identity and Access Management related requirements here. This can also cover the SSO federation, users and groups mapping, additional privileges which may be necessary to document

Example:

MYTestCustomer has an existing IAM setup based on AWS IAM. OCI IAM will be federated with AWS IAM.

The Oracle Database@AWS service will automatically create groups with administrative privileges, which will be assigned to the correct admins.

Additionally, all resources for these workloads will run in the same AWS organisation. There will be different accounts for the production, DR and non-production workloads.

3.2.5.2 Data Encryption / Key Management

Guide

Capture all the data encryption and key management requirements for the customer. Ensure that details related to current key usage (if any) and how it will be migrated for the Database@AWS deployment are documented here

Oracle ensures comprehensive data security with default encryption for data at rest and data in transit. Customers can manage their own encryption keys through Oracle Key Management for greater control. Oracle Data Safe enhances protection with features such as activity auditing, sensitive data discovery, user risk assessment and data masking. Additionally, Oracle Audit, Oracle Database Vault and Transparent Data Encryption (TDE) provide layered security controls to prevent unauthorised access and ensure compliance.

For Key Management, if customer-managed keys are required, then additional consideration needs to be given to the implication of choosing between the options:

- *Oracle Key Vault, running on AWS EC2. This option is normally positioned as the default “right” choice, unless there are specific reasons not to use this. Oracle Key Vault enables customers to securely manage security objects for Oracle Database@AWS in AWS, that includes encryption keys, Oracle wallets, Java Keystores (JKS), Java Cryptography Extension Keystores (JCEKS), and credential files with SSH private keys, etc.*
- *OCI Vault (running in OCI). Due to the additional latency between AWS and OCI, this should not be the default choice. However, for example in solutions where the keys need to be held in Hardware Security Modules (HSM), OCI Vault is a valid choice. (OCI subscription needed)*
- *AWS KMS (running in AWS). The integration with AWS KMS enables applications, AWS services, and databases on Exadata VM Clusters to leverage a single centralized key management solution. This solution also relies on Hardware Security Modules (HSM).*
- *File based. Unless this is a very simple solution, file based key management is not recommended.*

Example:

MYTestCustomer has a requirement to manage their own Oracle Database@AWS encryption keys. For this there are 3 options:

- Oracle Key Vault (OKV)
- OCI Vault
- AWS KMS

Oracle strongly recommends the use of OKV for the key management of Oracle Database@AWS, which is indeed what will be used for this deployment.

3.2.6 AWS / OCI Networking Requirements

Guide

Capture the Non-Functional Requirements for networking-related topics as it relates to how Oracle Database@AWS communicates with the rest of the AWS network.

- *Understand AWS Network topology – existing and new requirements.*
- *How should the application connect to the database (same VPC, peered VPC, Hub & Spoke topology, cloud WAN)?*
- *Connectivity from other locations (on premises, other AWS regions, other clouds, etc.)*

- Any requirement of using private endpoints
- DNS requirements (managed, or manually integrated)
- Latency requirements (both from AWS to Oracle Database@AWS, and if required, from OCI to Oracle Database@AWS).

If the OCI network elements of Oracle Database@AWS are to be leveraged in this workload, then standard OCI network requirements can also be described here, and you can use the networking questions in the [Annex](#)

Example:

The default implementation for MYTestCustomer is to use a Hub-and-Spoke model within AWS, leveraging a centralized Hub VPC for traffic inspection and management. However, for Oracle Database@AWS workloads, the applications and databases will be deployed in separate VPCs, directly peered to minimize latency. This approach ensures optimized performance for application-to-database traffic while maintaining a secure and scalable architecture. DNS for Oracle Database@AWS will be integrated with the existing AWS DNS environment.

3.2.7 Management and Monitoring

For the above section you can replace it accordingly with your requirements for Database@AWS project.

Example:

When you move or start an Oracle Database@AWS project, you have a choice to use the monitoring tools you are familiar with (should they support modern application architectures), including AWS, Oracle Enterprise Manager, or 3rd party tools. Because Oracle Database@AWS runs within an OCI tenancy, Oracle's recommended approach is to leverage native OCI observability services to ensure optimal performance and management.*

All the logs and monitoring metrics would be available through OCI, and AWS.

Oracle Database@AWS Observability within AWS

Oracle Database@AWS Observability within AWS

OCI native Observability services for Oracle Database@AWS:

Tool	Service	OCI Subscription	Notes
OCI Metrics	Basic monitoring	not required	Available by default
Database Management*	Advanced monitoring & admin	required	To be enabled - DBM Admin features only available in OCI
Ops Insights	Forecast & Capacity planning	required	To be enabled
Logging Analytics	Log analysis	required	OOTB Parsers/dashboards for Oracle DB

Oracle Database@AWS natively expose logs and maps basic metrics. To enable Administration and advanced analytics capabilities like Resource Forecast, OCI subscription is needed as the relevant services cannot be consumed against MACC. Once the Oracle Universal credit is available, it is possible to activate OCI native services as described here. OOTB dashboard and metrics are available on OCI tenant.

Oracle Enterprise Manager provides Database Management capabilities for customers who do not have an OCI subscription, offering a comprehensive solution for managing Oracle database.

Observability Tools for DB@AWS

Observability Tools for DB@AWS

Example:

MYTestCustomer wants to use AWS CloudWatch for Metrics, OCI Database Management, and Splunk for log analysis.

Tool	Task	Target	Location
OCI Database Management	Manage DB Instances + Monitoring and troubleshooting	All targets	OCI
AWS CloudWatch	Metric monitoring + notifications	All VM Clusters and DBs	AWS
Splunk	Log Analytics	All targets	Splunk

3.2.8 Automation & Integration (IaC)

Guide:

This subsection captures any requirements for integration into the customer's existing automation tools and practices - e.g. terraform, API, SDK requirements. Also, if the customer requires integration into automation platforms e.g. Jenkins, Ansible, GitHub, CD/CI AWS pipelines etc. these should be recorded.

Example:

At the time of this document creation, no Automation requirements have been specified.

MYTestCustomer can leverage AWS and OCI Cloud Native automation tools to provision end to end Exadata Infrastructure, Cluster VM and the database. The following tools are available:

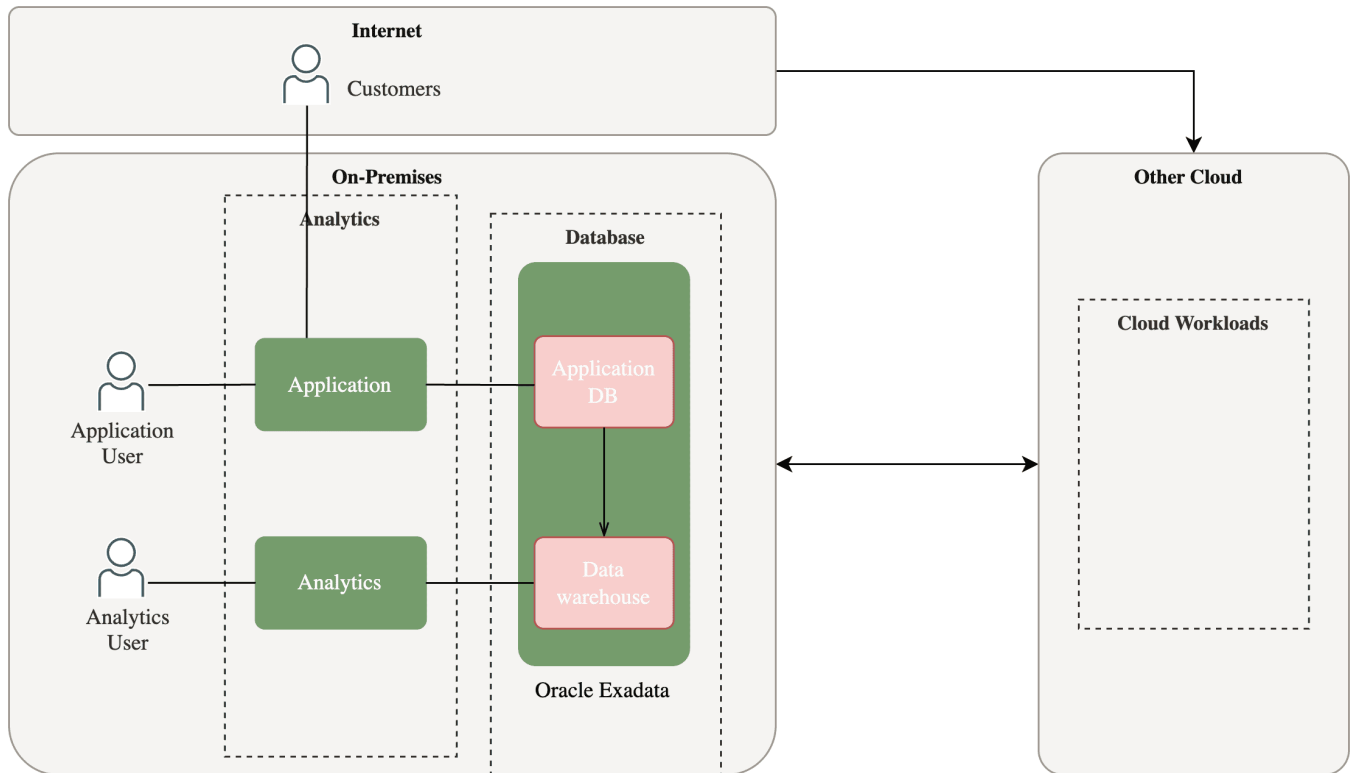
AWS	OCI
AWS CLI	OCI CLI
REST API/SDK	REST API/SDK
CloudFormation	Ansible
Terraform	Terraform

3.3 Current State Architecture

Guide: Try to capture as much information as possible and suitable for the context of this project. The logical diagram below can be extended or modified for the actual customer requirement. E.g. there can be partners instead of just customer, there can be other cloud vendors already in the picture and so on. Below diagrams draw.io file is available in the diagrams repository for further editing. You can go as broader as you like but capture as much details as possible for the workloads in consideration.

Example:

The current implementation of this workload runs fully in MYTestCustomer's own data center. 1 x Exadata Machine contains all database workloads, and both the application and the BI environment are running in the same network. Internet-based customers have access to the self-service part of the application.



Current State Logical Architecture

3.4 Future State Architecture

Guide:

The Future State Architecture can be described in various forms. In the easiest case, we describe a Logical Architecture, possibly with a System Context Diagram. A high-level physical architecture is mandatory as a description of your solution.

This should be the final architecture as part of the pre-sales solution, not an intermediate or draft version

Additional architectures, in the subsections, can be used to describe needs for specific solution.

3.4.1 Solution Best Practices

Guide:

Oracle highly recommends following the solution best practices for AWS as well for OCI. More details on planning your solution at Oracle Database@AWS can be found [here](#)

Oracle highly recommends following best practices to protect your Database@AWS solution on OCI:

Mandatory Security Best Practices:

Guide:

Oracle Database@AWS is deployed within an AWS framework, and therefore any AWS security best practices apply to the whole solution / workload. Those best practices are not discussed further in this document.

Oracle Database@AWS is also part of an OCI tenancy, and as such OCI Security best practices also apply and are described here:

AWS Security Best Practices:

Guide:

Describe any AWS best practices, security it critical with data encryption, VPC Service Controls. Network design should use subnets, firewall rules and secure communications such as VPC peering.

Naming Conventions:

Guide:

For Oracle Database@AWS workloads without any additional OCI components, it is acceptable to refer to existing AWS naming conventions, rather than describing AWS naming conventions here.

This chapter describes naming convention best practices and usually does not require any changes. If changes are required, please refer to [Landing Zone GitHub](#). The naming convention zone needs to be described in the Solution Design by the service provider.

Use this template ONLY for new cloud deployments and remove it for brownfield deployments. A naming convention is an important part of any deployment to ensure consistency, governance, and security within your tenancy. Find here Oracle's recommended best practices.

OCI Landing Zones:

Guide:

This chapter describes landing zone best practices and usually does not require any changes. If changes are required please refer to [Landing Zone GitHub](#). The full landing zone needs to be described in the Solution Design by the service provider.

Use this template ONLY for new cloud deployments and remove it for brownfield deployments. When Oracle Database@AWS is implemented, it is always linked to an OCI tenancy.

Note also that there are 2 types of Oracle Database@AWS environment. Those environments where the OCI tenancy is used for other services should design an OCI Landing Zone according to the normal considerations for those workloads. For those environments where the OCI tenancy is only used to support the Oracle Database @ AWS, a minimal landing zone can be used. In that case we still need to consider items such as:

- *Compartments (automatically created by the Oracle Database@AWS service)*
- *IAM (groups and policies created as part of the Oracle Database@AWS service)*
- *Security posture (are services like Cloud Guard, Security Zones, Vulnerability Scanning needed on OCI)*
- *Network Structure (VCNs automatically created by the Oracle Database@AWS service)*
- *Network Security*
- *Network Connectivity*
- *DNS*

Naming conventions:**3.4.2 Logical Architecture**

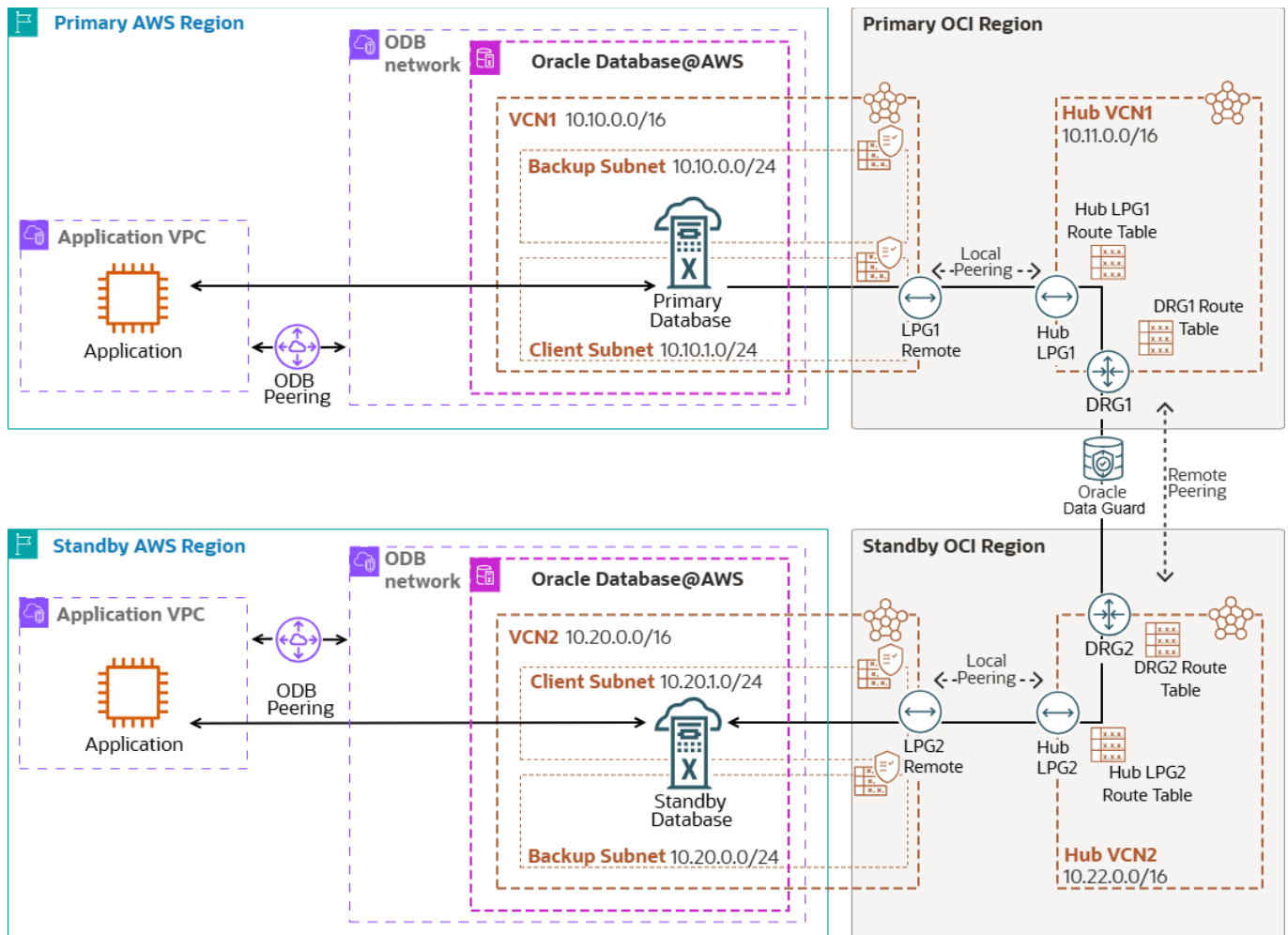
Guide:

Provide a high-level logical Oracle solution for the complete solution. Indicate Oracle products as abstract groups, and not as physical detailed instances. Create an architecture diagram following the latest notation and describe the solution.

It is expected to provide a full Multicloud architecture in this section, including all AWS elements which have direct or indirect access to Oracle Database@AWS.

To implement a solution the Physical Architecture is needed in the next chapter. The physical notation can show individual components with physical attributes such as IP addresses, hostnames, or sizes.

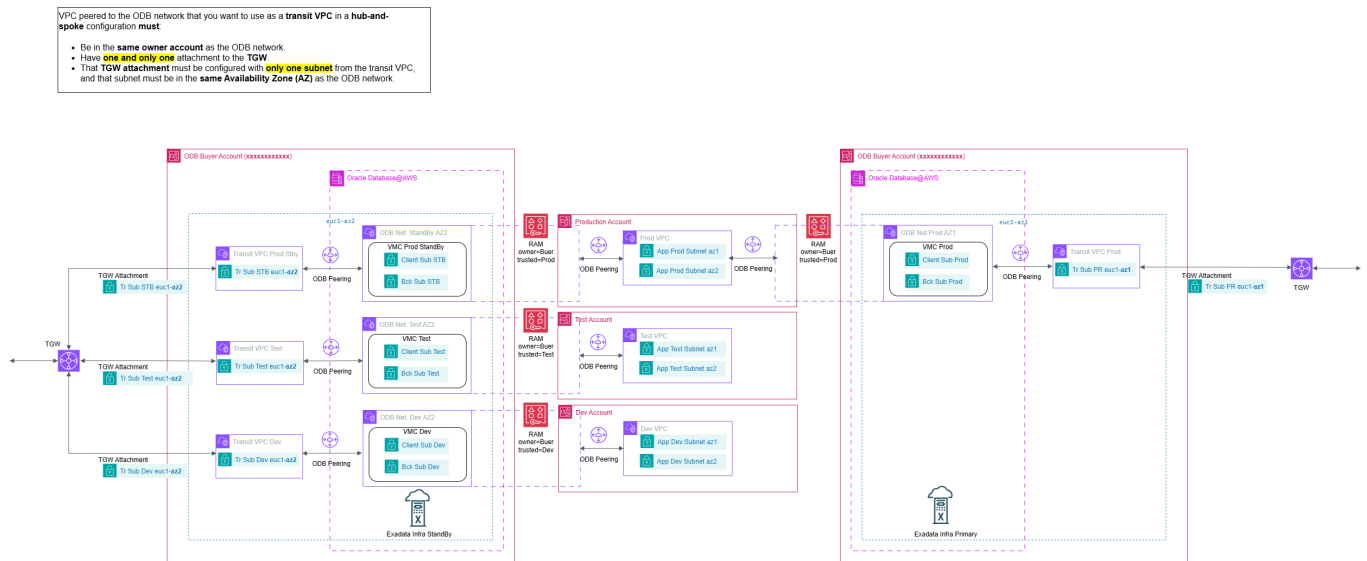
The Oracle Cloud Notation, OCI Architecture Diagram Toolkits



DB@AWS Physical Across Regions

Examples:

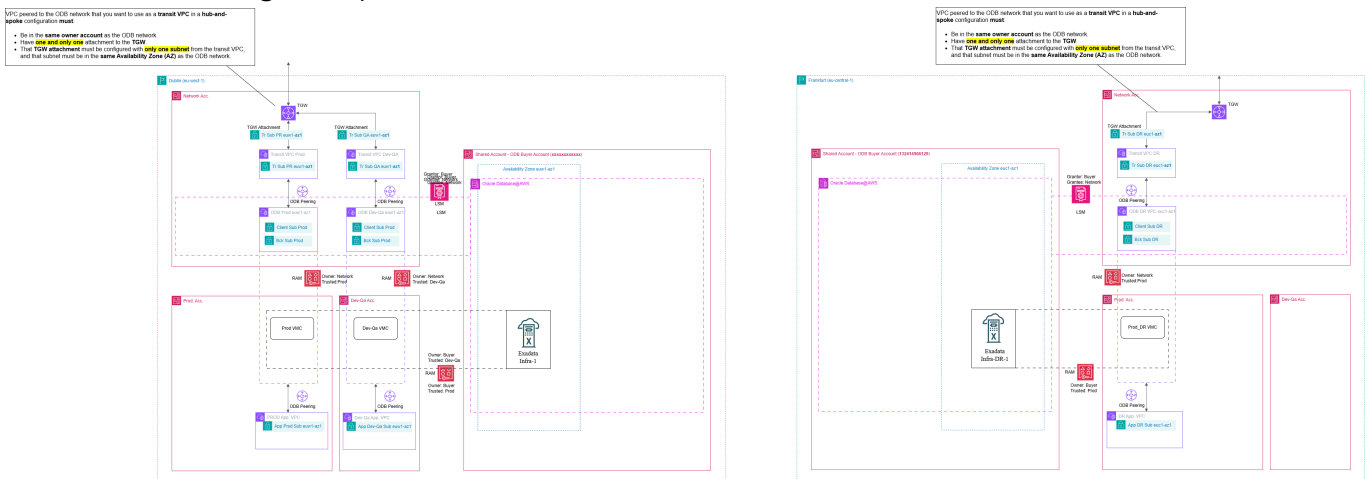
1. DB@AWS target architecture with ODB network segmentation across three environments plus DR
 - Internal network traffic is routed through ODB peering, while external network traffic uses a TGW hub-and-spoke configuration.
 - MAA Cross-AZ setup.



ODB network segmentation across three environments plus DR

2. DB@AWS target architecture with account segregation. [\(Drawio source file\)](#)

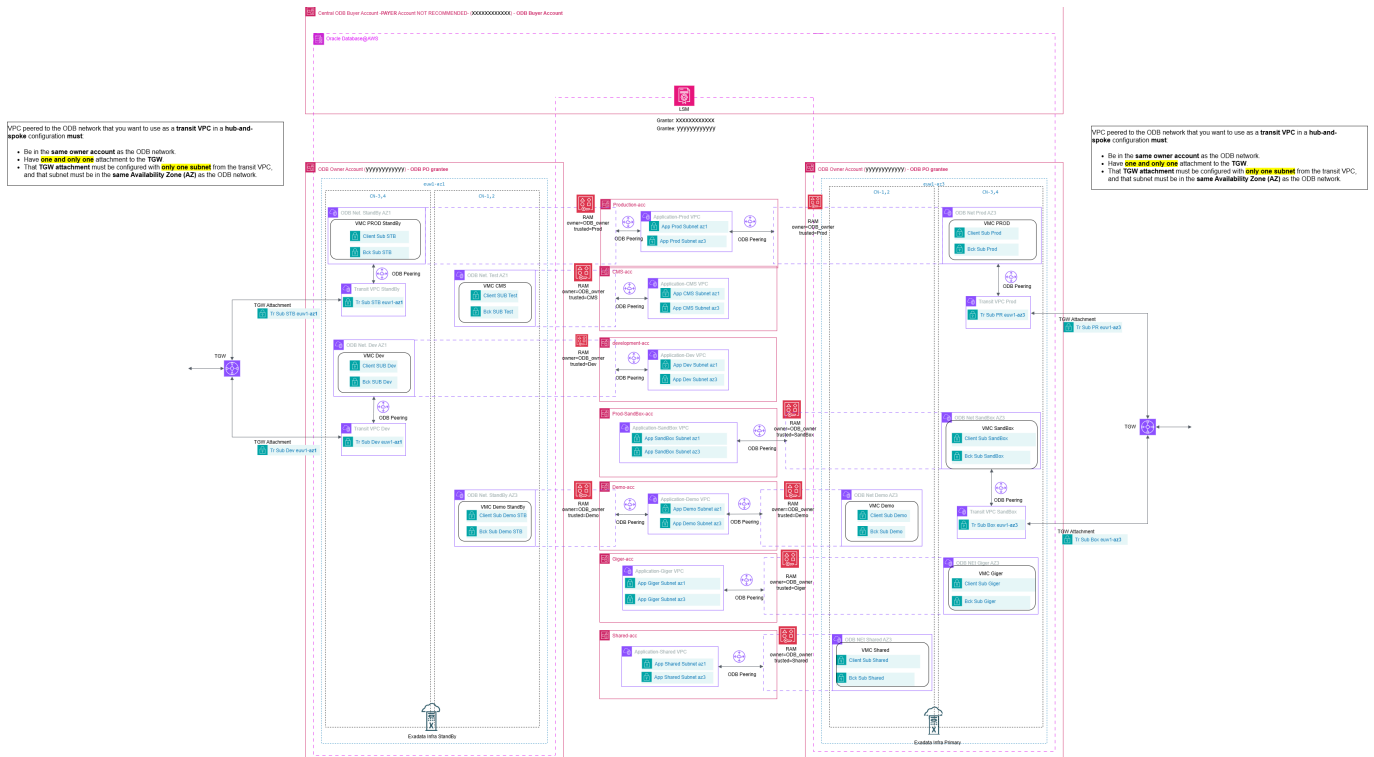
- The AWS Network Account owns all network components, including the ODB networks and Transit VPCs.
- The AWS PROD Account owns the PROD VM Cluster and the AWS TEST Account owns the TEST VM Cluster.
- ODB network segmentation covers two environments plus DR, with internal traffic routed through ODB peering and external traffic through a TGW hub-and-spoke configuration.
- MAA Cross-Region setup.



DB@AWS target architecture with account segregation, AWS Network Account owns all network components

3. DB@AWS target architecture with ODB network segmentation across seven environments plus DR. [\(Drawio source file\)](#)

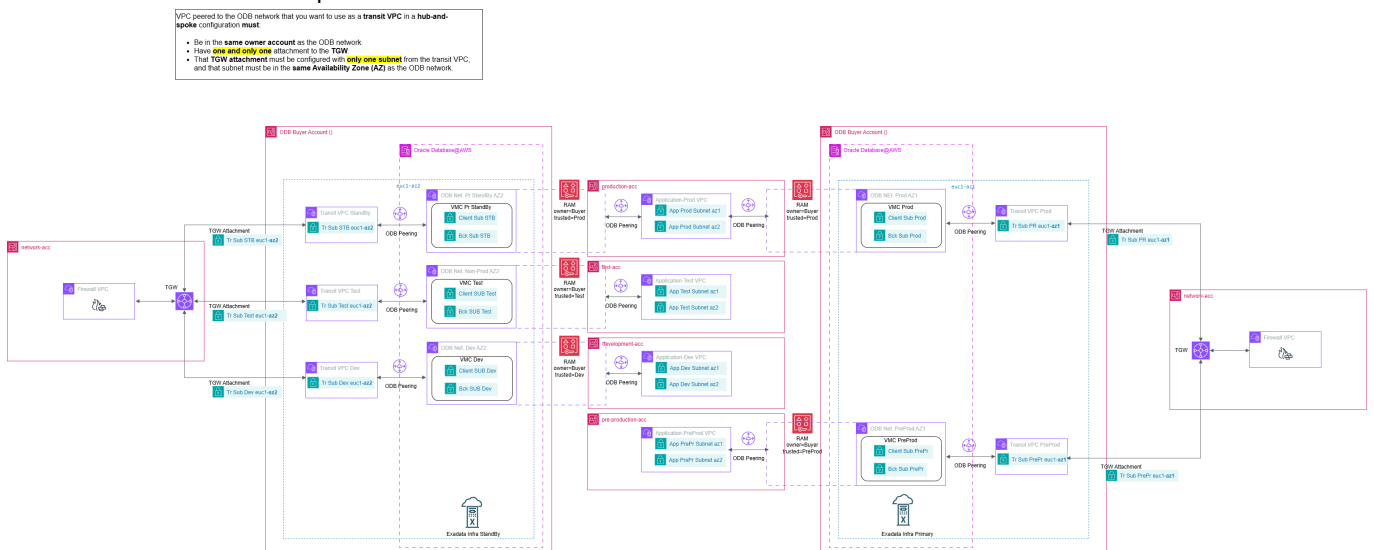
- Internal network traffic is routed through ODB peering, while external traffic uses a TGW hub-and-spoke configuration.
- ADB-D consolidation requires two additional CNs (Compute Nodes) in the Exadata Basic configuration to provide enough local storage to host six AVMCs (Autonomous VM Clusters), with five ACDBs (Autonomous Container Databases) each.
- MAA Cross-AZ setup.



DB@AWS target architecture with ODB network segmentation across seven environments plus DR

4. DB@AWS target architecture with ODB network segmentation across four environments plus DR. ([Drawio source file](#))

- Internal network traffic is routed through ODB peering, while external traffic uses a TGW hub-and-spoke configuration plus a Firewall VPC.
- MAA Cross-AZ setup.

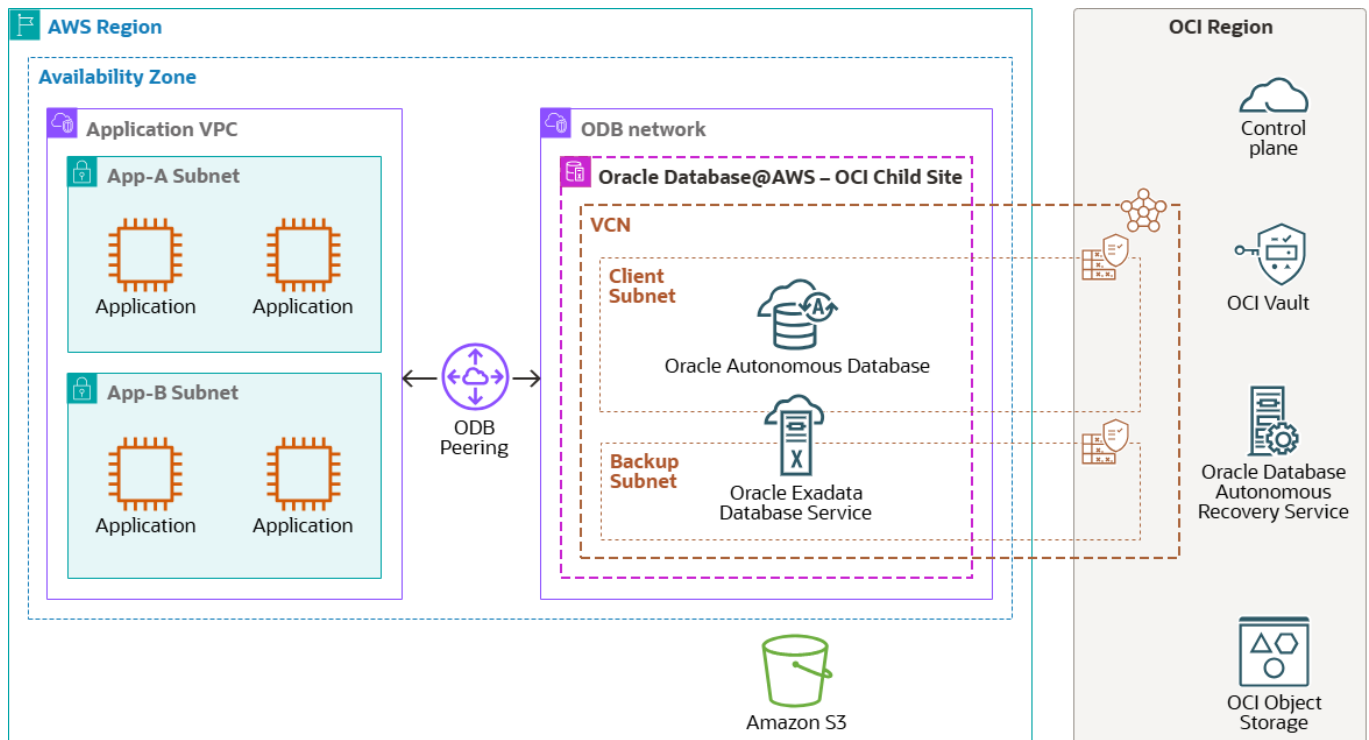


DB@AWS target architecture with ODB network segmentation across four environments plus DR.

3.4.2.1 Database Consolidation with Multiple Service Levels

Based on requirements for data residency and local application performance this solution delivers Database service in AWS data centers using ExaDB-D. Considering Oracle Maximum Availability Architecture (MAA) multiple service levels have been defined in the logical solution architecture below. Also note that data databases with similar requirements i.e. security posture, performance attributes, DB version etc. can be combined into a DB VM Cluster under ExaDB-D@AWS.

Use the below diagram as a baseline to document the deployment specific details. This diagram can be edited to incorporate other components, systems and applications hosted on AWS or OCI to cover. Also, we are listing here the ExaDB-D and ADB-S but you can keep either or both as per customer requirements.



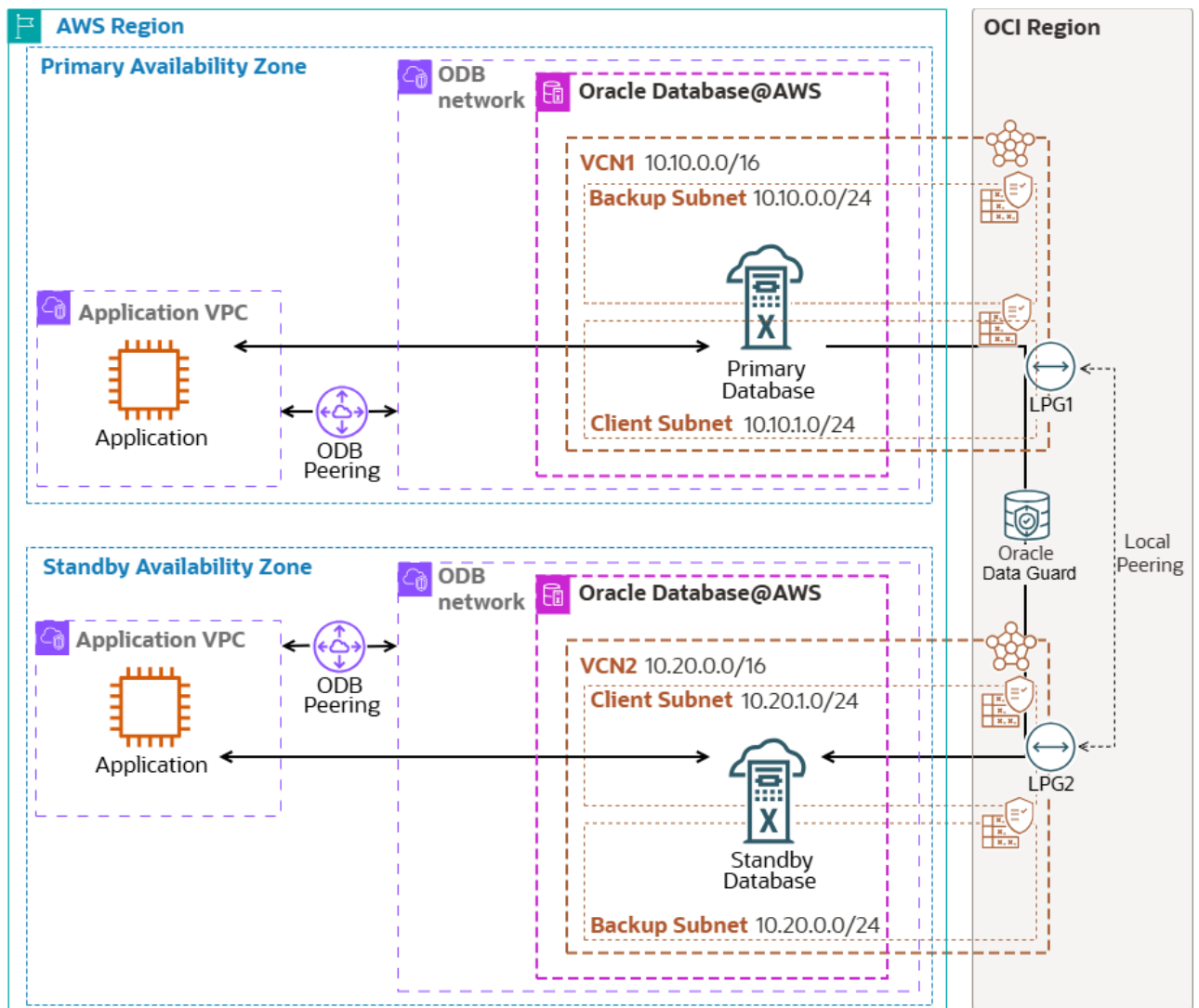
DB@AWS High Level Deployment

The physical architectures can have multiple service levels for *Production Database Service A (Standby in same AWS region), and Production Database service B (Standby across AWS regions)

Please note that standby in same AWS region is a work in progress at the moment.

3.4.2.2 Service Level with High Availability, Disaster Recovery and Data Protection – within region

Production Database Service A Using MAA Reference Architecture “Gold” for Production Database Service A which has High Availability with Oracle RAC, Disaster Recovery with Oracle Data Guard and Data Protection with copies in both regions with Oracle Zero Data Loss Recovery Service.



DB@AWS MAA Gold Same Region

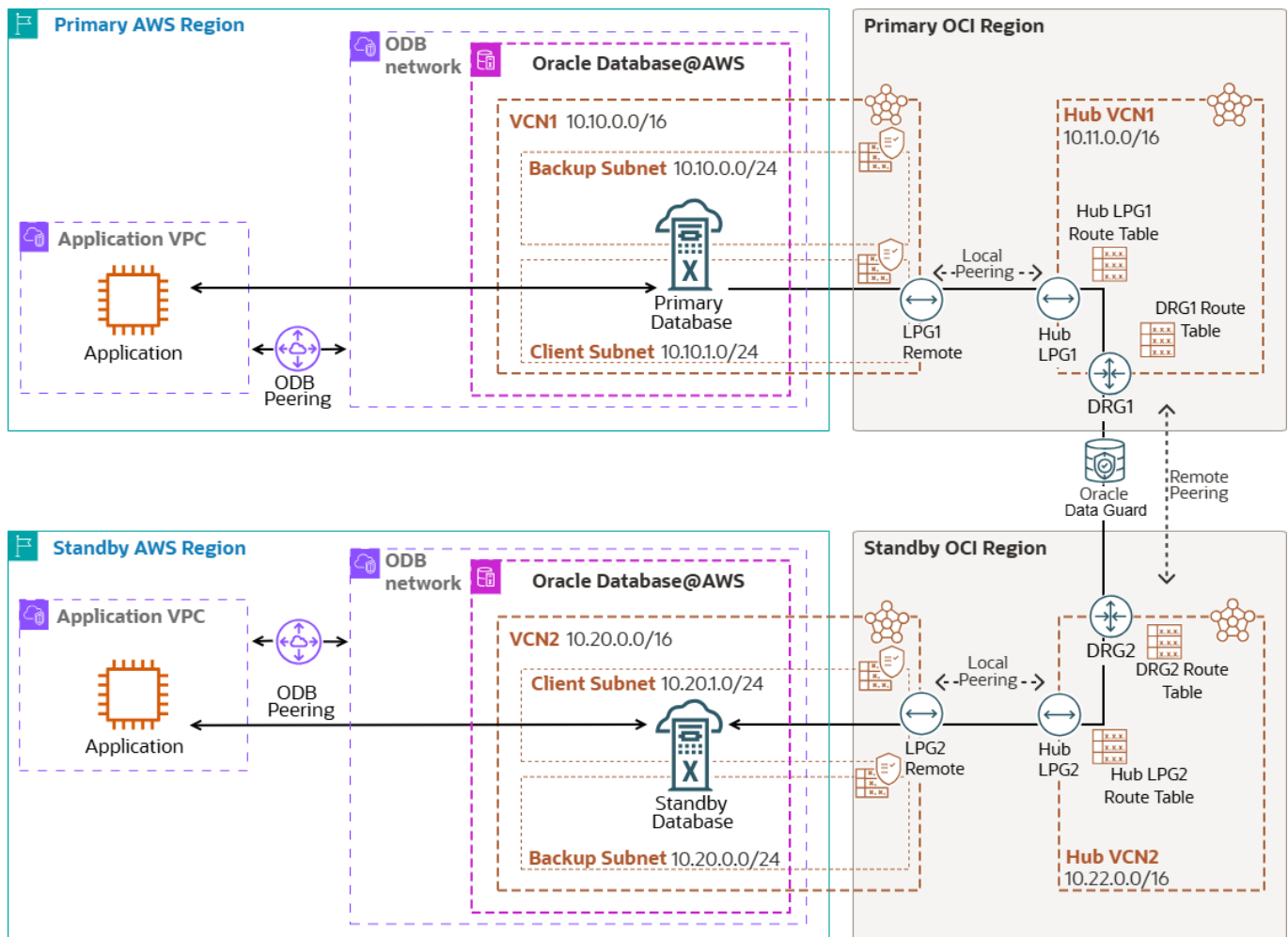
Details of Database Service A based on MAA Reference Architecture "Gold" in same region (Work in Progress)

3.4.2.3 Service Level with High Availability, Disaster Recovery and Data Protection – across regions

Production Database Service B

Using MAA Reference Architecture "Silver" for B and C since they are identical but split across two data centers for efficient use of available resources. Non-Production Database Service B and C has High Availability with Oracle RAC and Data Protection with copies in both data centers with Oracle Zero Data Loss Autonomous Recovery Service (ZRCV) and ZRCV Replication.

Additional information on MAA Reference Architectures can be found here: [Oracle MAA For Oracle Database@AWS](#)



Details of Database Service B based on MAA Reference Architecture “Gold” across regions

3.4.3 Physical Architecture

Guide:

The solution architecture is typically described in a physical form. This should include all solution components. You do not have to provide solution build or deployment details such as IP addresses.

It is expected to provide a full Multicloud architecture in this section, including all AWS elements which have direct or indirect access to Oracle Database@AWS.

Please describe the solution with an architecture image plus a written text. If you have certain specifics you like to explain, you can also use the Solution Consideration chapter to describe the details there.

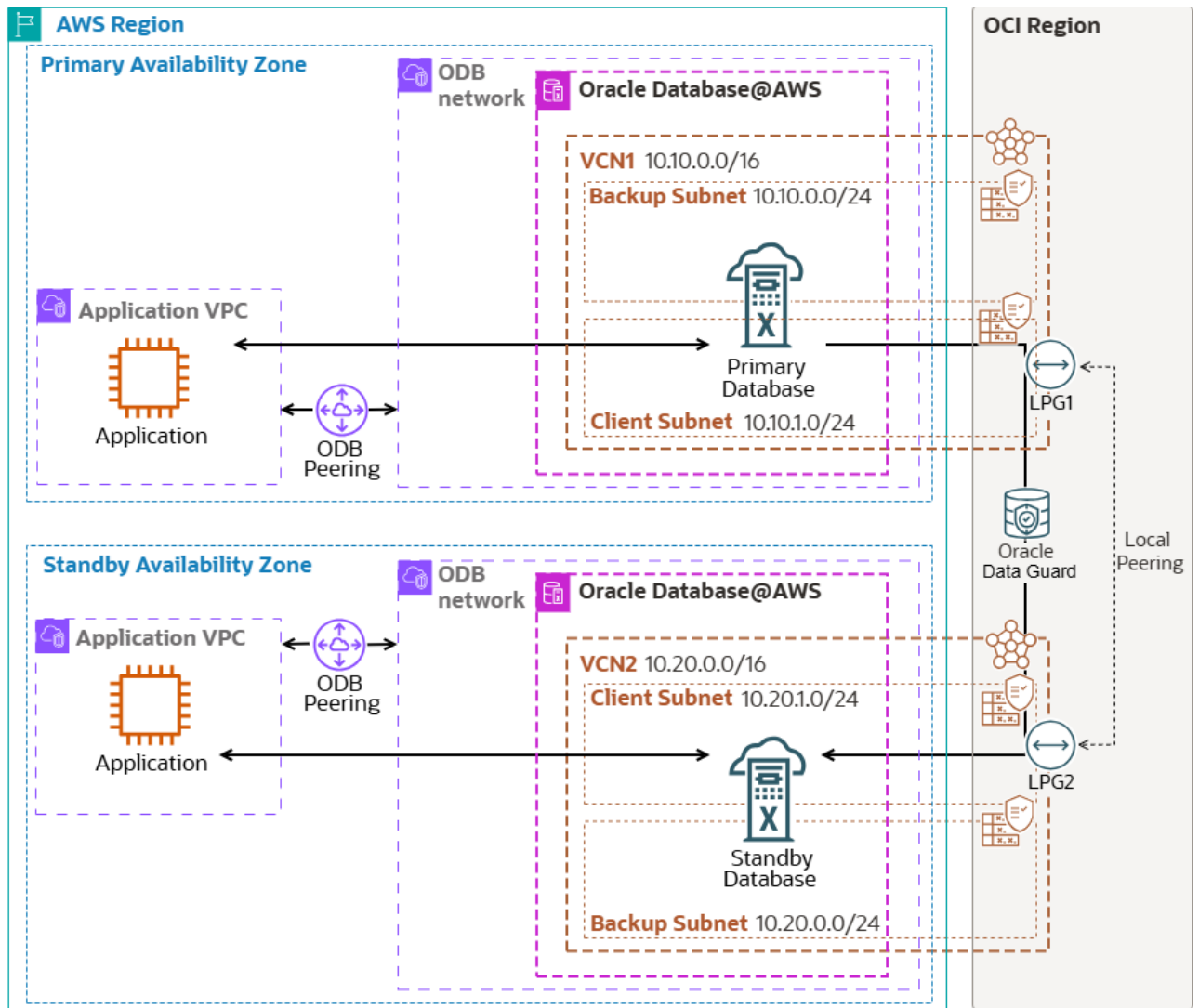
[The Oracle Cloud Notation, OCI Architecture Diagram Toolkits](#)

Reference:

[StarterPacks \(use the search\)](#)

Example:

Physical Architecture of Production and DR/Standby – same region

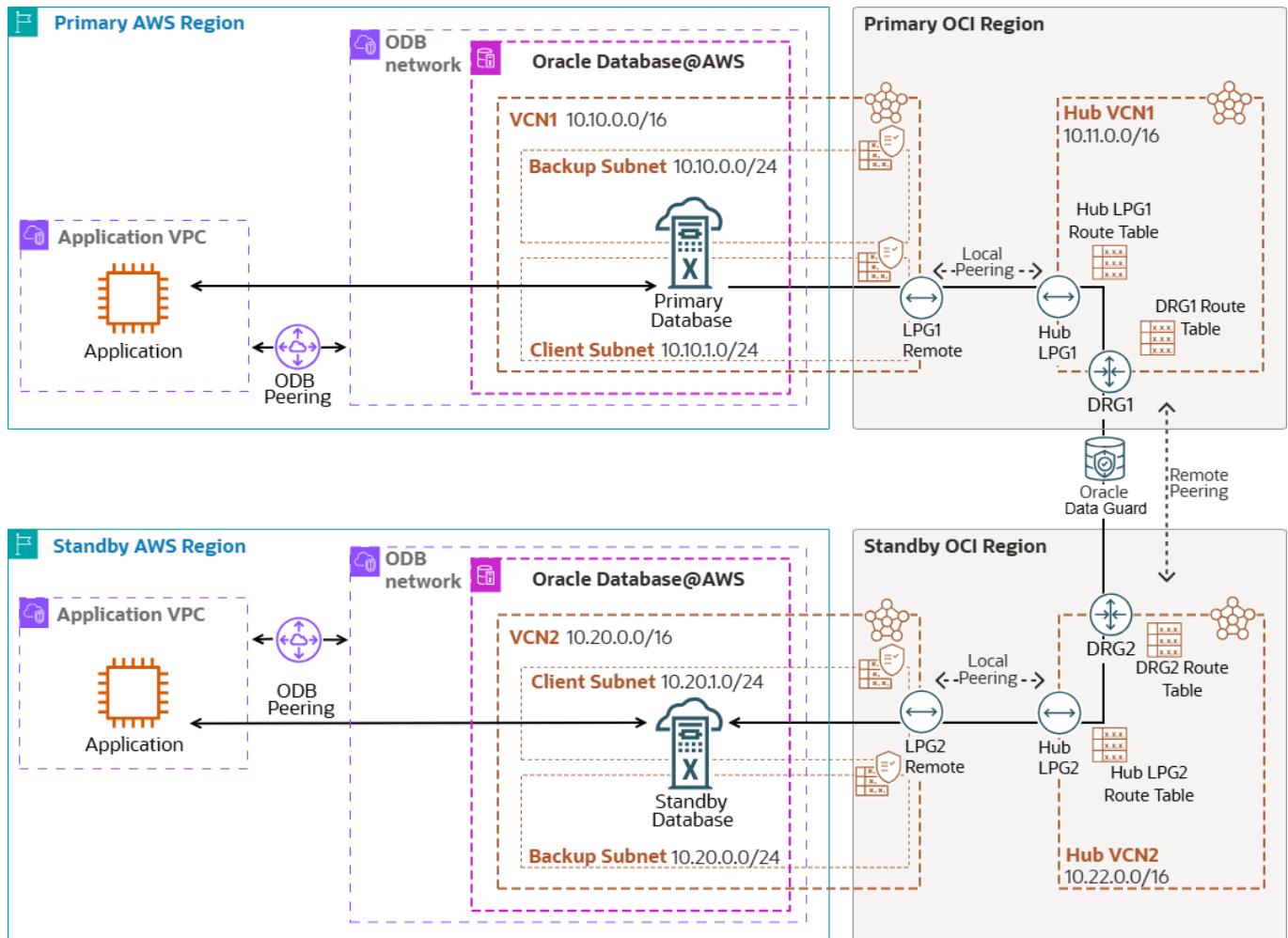


DB@AWS MAA Gold Same Region

Physical Architecture of Production and DR/Standby – across regions

Guide:

Please use below diagram and provided example text to write further according to your customer requirements.



DB@AWS Physical Across Regions

In this example there are only 3 VM clusters in use to represent three different application DBs. VM Cluster segregation can be based on environment, DB versions, licensing type, security requirements or other attributes as well. When there are multiple workloads across multiple VM clusters, then additionally notes can be made on VM Clusters fulfilling the following requirements:

- Network segmentation due to security:

Each VM Cluster can be placed in its own AWS VPC or OCI Subnet. However this is not mandatory and multiple VM Clusters can co-exist within single VPC and Subnet as long as enough IP Addresses are available. This is what above diagram shows but this can be further enriched with multiple client and backup subnets.

- Workload isolation

Each VM Cluster have its own dedicated resources in CPU, Memory and Storage Capacity.

- Application / User separation

Each VM Cluster can have its own users and groups as defined in the Identity Service.

- Required Service Level Objectives (for example MAA Gold, Silver)

Each VM Cluster is designed against a given MAA Reference Architecture.

High Availability will be granted with Oracle RAC features.

Information on Oracle RAC Best Practices for DB 19c is here: [Overview of Oracle RAC and Clusterware Best Practices](#)

Additional information on RAC can be found in the MAA Reference Architectures here: [Oracle MAA Reference Architectures](#)

Disaster Recovery capabilities will be granted with Oracle Data Guard. Information on Oracle Data Guard Best Practices for DB 19c is here: [Oracle Data Guard Best Practices](#)

Additional information on Data Guard can be found in the MAA Reference Architectures here: [Oracle MAA Reference Architectures](#)

In case of unplanned/planned downtime of the production site, an automated or manual mechanism will allow failover/switchover to the other production platform.

Oracle Transparent Data Encryption (TDE) will be used. This is mandatory with ExaDB-D@AWS

The Control Plane of this ExaDB-D@AWS is in the local OCI Public Cloud region. The control plane is a sophisticated cloud software suite for order management, subscription billing, identity management, UI and REST API services, network connectivity, security controls, infrastructure and lifecycle management. Oracle Identity and Access Management (optionally federated with AWS IAM), lets you control who has access to your Exadata resources. One can control what type of access a group of users have and to which specific resources.

For data protection all VM Clusters on the same Exadata Infrastructure share the same Backup Subnet.

3.4.4 Key Solution Proposal Components

Example:

Our proposal is based on the following main technical components:

3.4.4.1 Exadata Dedicated@AWS (ExaDB-D@AWS)

Oracle ExaDB-D@AWS brings the performance, automation, and economics of Exadata Database Service and Autonomous Database fully managed by Oracle to data centers in AWS. This is the easiest way for MYTestCustomer to start using cloud database resources within AWS while continuing their deployment of AWS. ExaDB-D incorporates unique optimizations that enable databases to run faster with less management and at lower costs, to better leverage MYTestCustomer data.

Value Proposition

Oracle's value proposition to support MYTestCustomer's needs is built around the Oracle ExaDB-D cloud platform:

Resilience and Availability

- Modern, highly available, and secure data architecture by design
- Simplified patch maintenance and proactive threat management
- Through ExaDB-D@AWS, these services can be used inside an AWS datacenter, natively part of the AWS network.

Performance, agility, and flexibility

- Low latency to keep data as close as possible to applications and choose the timeline for modernizing these databases as needed
- Ability to scale the platform as needs change without increasing complexity
- Maximum availability architecture (e.g. PRA with fast RTO and RPO, service continuity, etc.)
- Obsolescence and risks
- Simplify and standardize to enhance security
- Automation to achieve more predictable service levels and address a wide range of security vulnerabilities Security

- Encryption of all data at rest and in transit.

Speed of implementation / Time to Market

- Time to Market optimization is made possible thanks to the comprehensive "Infrastructure as Code" approach, which allows all environments to be managed through a DevOps methodology to ensure repeatability.
- Implementation of applications directly and without the need for infrastructure teams, greatly accelerating their implementation and simplifying their maintenance.

Environmental Impact

- Achieving carbon neutrality in 2025
- Reduce CO2 production by 50% in 2030
- Reaching Net Zero in 2050
- Efficiency of Hardware Oracle OCI ExaDB-C@C solutions: 4d eco-design program, certifications, etc.
- Renewable energy use for all global operations by 2025
- End-to-end infrastructure reuse by our engineers: 4d eco-design program, supply chain redesign for delivery and recovery with 99.9% reuse/recycling rate in 2022
- Reduction in the average measured carbon footprint (CSR audit firm) of our customers migrating to our cloud solutions in Europe: 40%
- Align with Sustainable Development Goals
- Aware of climate issues, Oracle has set targets to reduce CO2 production, both internally at the group level and for all of its partners.

3.5 Solution Considerations

Guide:

Describe certain aspects of your solution in detail. What are the security, resilience, networking, and operations decisions you have taken that are important for your customer?

Typical key considerations are:

- *High Availability and Disaster Recovery*
- *Security*
- *Networking*
- *Manageability and Observability*

Feel free to describe other aspects and important requirements for your customer, and how your solution addresses them in this chapter.

3.5.1 Oracle Database@AWS High Availability and Disaster Recovery

Guide:

Proposed to be HA/DR/MAA design and details

- *Resilience on OCI*
- *Workload Related Content*
- *What happens when site 1 is down?*
- *What happens when XYZ happens?*
- *How is HA achieved?*

Due to the Multicloud nature of the Oracle Database@AWS solution, there can be considerable traffic between AWS and OCI datacenters in Disaster Recovery scenarios. For example:

By default, Oracle Data Guard redo logs are shipped from the primary database to the standby database via the managed Dynamic Routing Gateway (DRG), directing traffic through OCI's network infrastructure. Detailed architecture is available [in Oracle documentation](#).

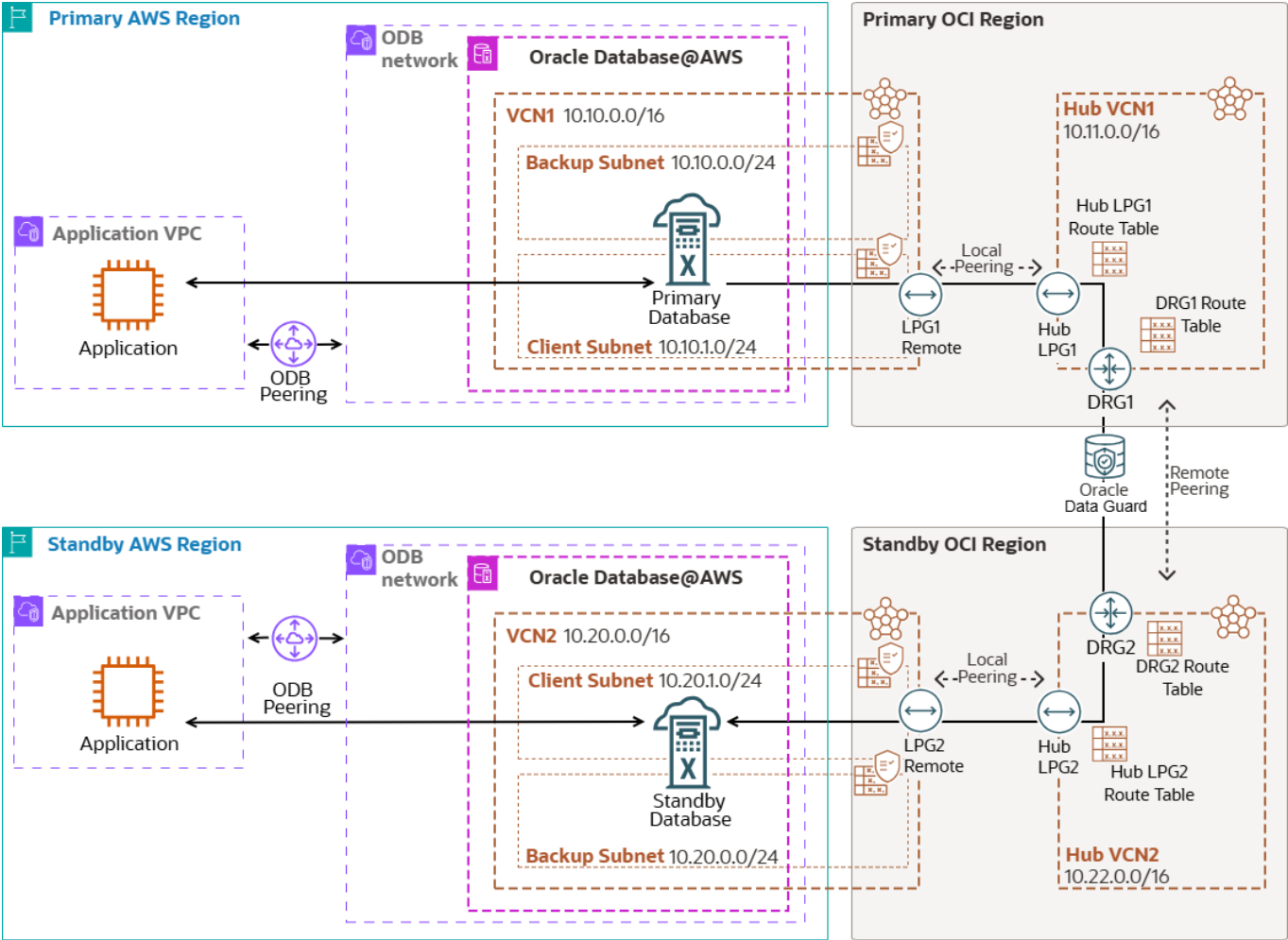
- If the standby database is hosted in the same Oracle Database@AWS region:
 - Traffic can be routed over AWS VPC or through OCI Local Peering Gateway (LPG).
- If the standby database is hosted in a different Oracle Database@AWS region:
 - Data Guard traffic can be routed via AWS's VPC, but can also be routed via Oracle Cloud backbone using the following architecture: <https://docs.oracle.com/en/solutions/dr-local-regional-standby-db-at-aws/index.html#GUID-20F21093-AF9B-421D-A0B4-CA87155F6337> (both options incur cross-region data transfer cost)

Example:

MYTestCustomer requires to have a standby in a separate region.

Following has been decided

- Production Databases:
 - Database Service A
 - Full RAC
 - DataGuard with region 1 as primary and region 2 as standby
 - Backups will be done from the local instance to the local backup environment



Oracle Database@AWS Multi-Region DR with Oracle Data Guard

3.5.2 Security

Guide:

Please describe your solution from a security point of view. Generic security guidelines are in the [Annex](#)

Note that Key Management is always something that needs to be thought of clearly for Oracle Database@AWS Exadata deployments. As per section [Security Requirements](#), if customer-managed keys are required, then Oracle Key Vault is usually the default correct choice. However, if for example keys need to be held in hardware, then OCI Vault or AWS KMS should be used.

Example:

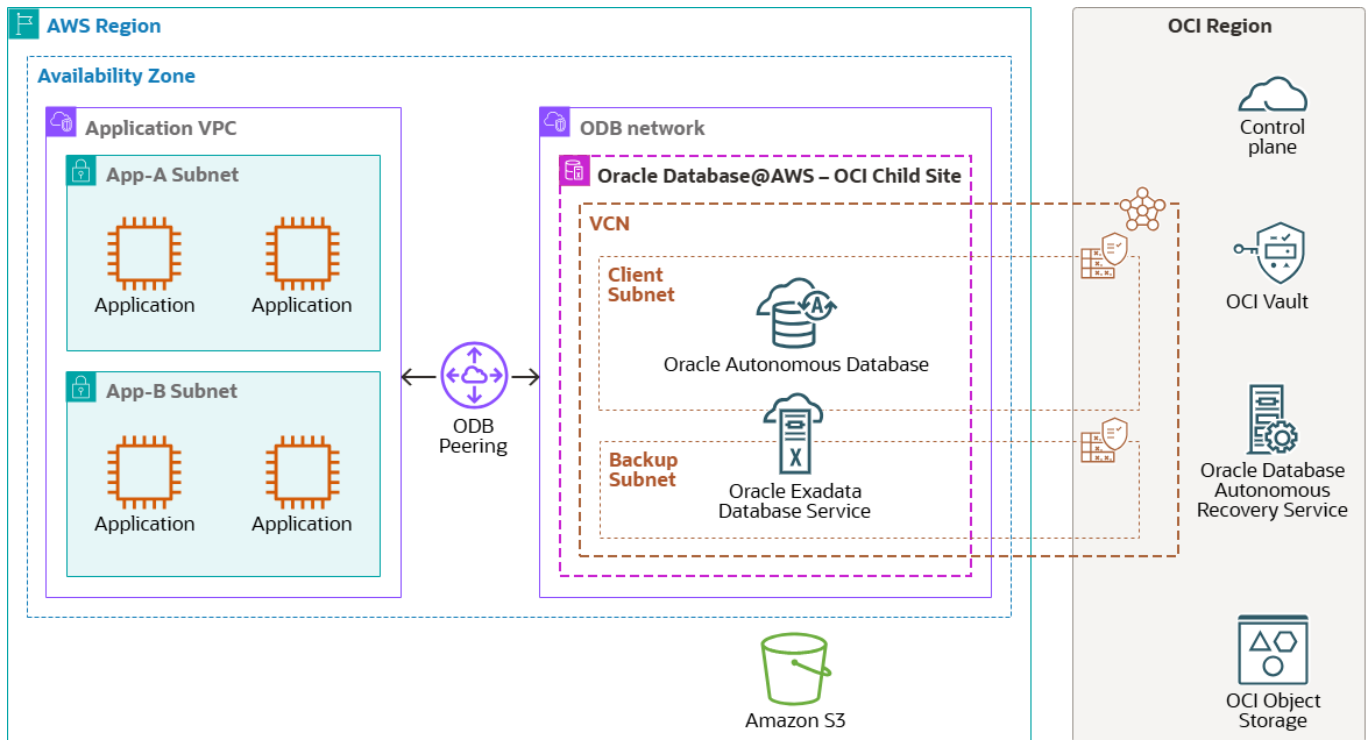
Key management options for Oracle Database@AWS can be based on following options.

Key Management Option	Service	Type	Further Information
Oracle Managed Keys	Oracle managed vault	Default in Exadata Cloud Infrastructure and Autonomous DB	Oracle manages the encryption keys

Key Management Option	Service	Type	Further Information
Customer Managed Keys	OCI Vault (service in OCI, UCM credits needed)	Virtual Vault Multitenant partition	HSM Keys - A master encryption key protected by an HSM is stored on an HSM and cannot be exported from the HSM. All cryptographic operations involving the key also happen on the HSM. These keys are FIPS Level 3 compliant.
Customer Managed Keys	OCI Vault (service in OCI, UCM credits needed)	Private Vault Own partition in HSM, Better and consistent transactions per second for cryptographic operations. Cross Region Replication. Cross Region Back/Restore of Keys	Software Keys - A master encryption key protected by software is stored on a server and can be exported from the server to perform cryptographic operations on the client instead of on the server. While at rest, the software-protected key is encrypted by a root key on the HSM. These keys are FIPS Level 1 compliant.
Customer Managed Keys	Oracle Key Vault (license needed)	Oracle Key Vault running on OCI, Customer on-premise or in AWS EC2.	Software on VM, manual installation. Located on AWS, "local" to the database.
Customer Managed Keys	AWS KMS	Key material origin = KMS	By default, a KMS key is created using the standard AWS KMS HSM. This HSM type can be thought of as a multi-tenant fleet of HSMs that allows for the most scalable, lowest cost and easiest key store to manage from customer's perspective.
Customer Managed Keys	AWS KMS	Key material origin = AWS CloudHSM	While a AWS CloudHSM cluster is hosted by AWS, it is a single-tenant solution that is directly managed and operated by the customer. The customer owns much of the availability and performance of KMS keys in a AWS CloudHSM cluster.

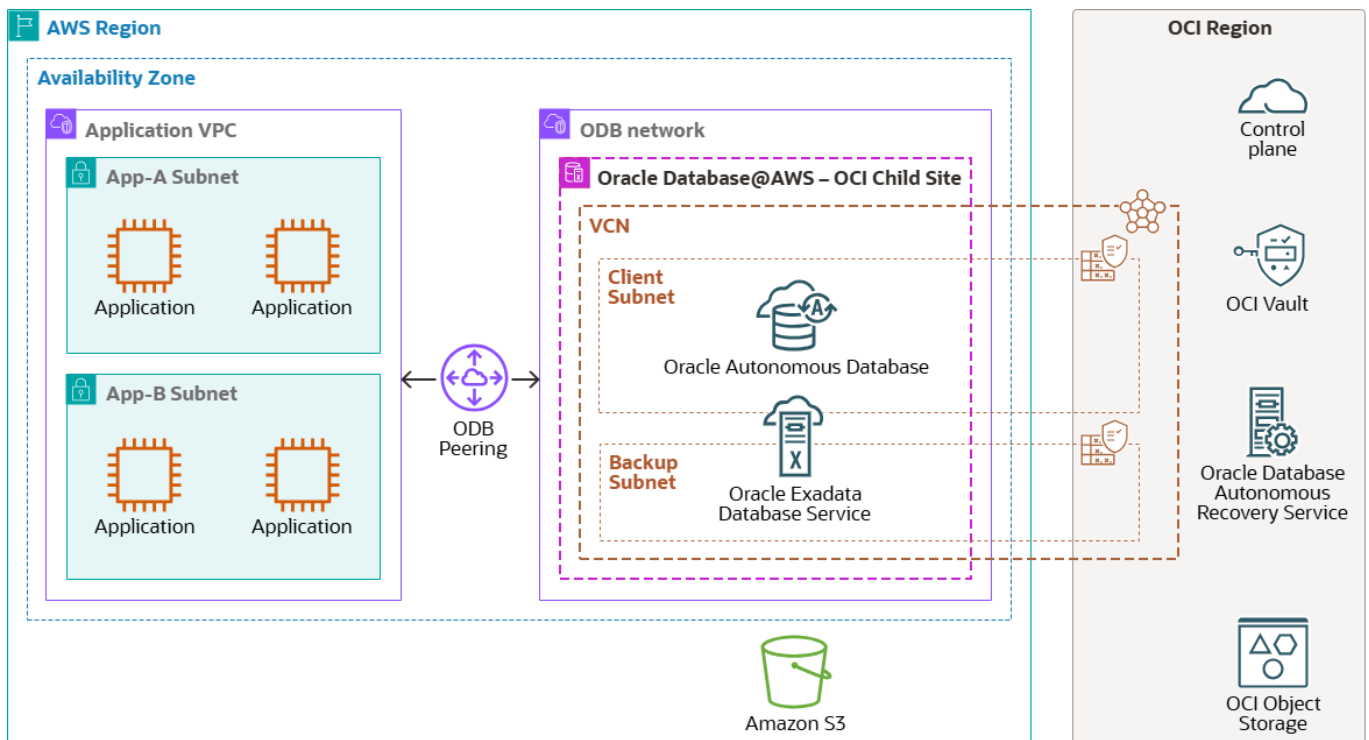
Note: Cross-region Data Guard and restoring databases to a different region are currently not supported for databases that use AWS KMS customer managed keys for key management.

--- To be replaced with a more detailed diagram ---



Oracle Database@AWS with OKV on AWS VMs

--- To be replaced with a more detailed diagram ---



Oracle Database@AWS with OCI Vault

3.5.3 Networking

Example:

Below are standard OCI guidelines and examples for networking. This section also has Database@AWS specification to be considered

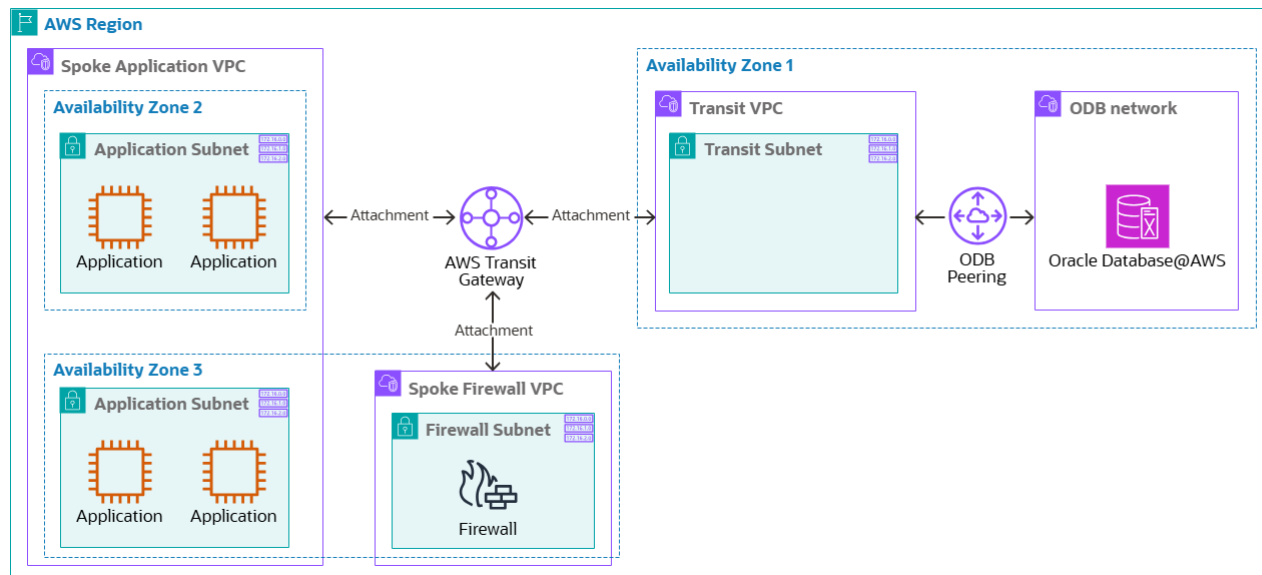
Networking Example

Guide: Please describe your solution from an Oracle Database@AWS networking design point of view.

Example 1:

Hub and Spoke Topology

MYTestCustomer would like to use their existing Hub and Spoke network topology to route all the traffic through their Network Virtual Appliance/Firewall. Applications and DB@AWS resources can reside in separate VPCs for increase network isolation and security.



Oracle Database@AWS Hub and Spoke Network Topology

Example 2:

Network Peering Topology

VPC peering topology connects your applications in one VPC to the database in another VPC. VPC peering connects two VPCs so that resources in each network can communicate with each other in the same project, a different project of the same organization, or even different projects of different organizations.

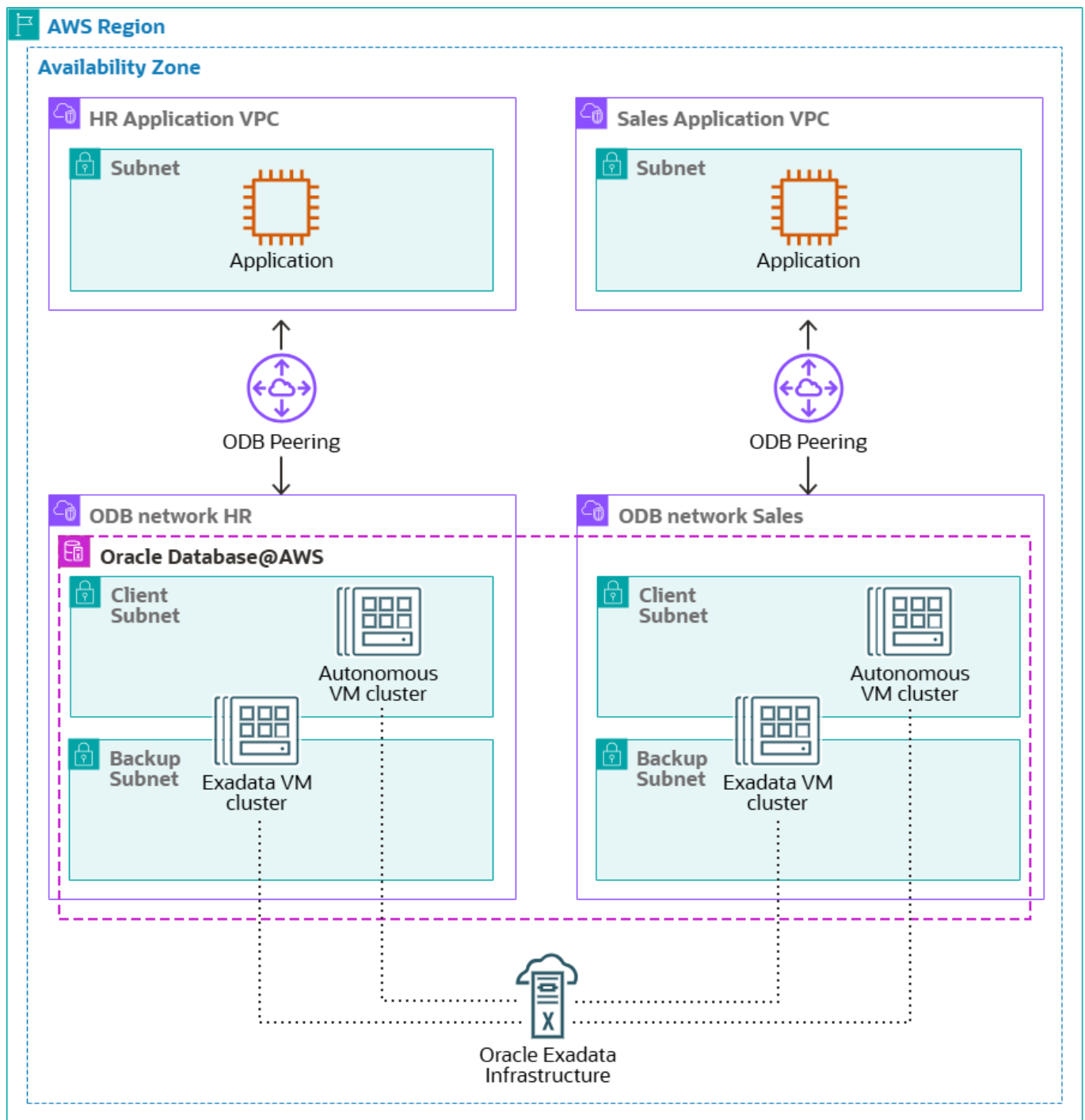
--- Need to create this diagram --- Oracle Database@AWS Network Peering Topology

Example 3:

Multiple VPC Topology

If you want to isolate database workloads between two lines of businesses, use the multiple VPC topology. A Sales and Marketing team can have their own VPC and subnets for their databases in a VM Cluster inside an Oracle Exadata Database Service infrastructure.

To isolate workloads on Oracle Exadata Database Service at the VM Cluster level, deploy multiple clusters in isolated VPCs.



Oracle Database@AWS Multiple Networks

Shared VPC Topology

--- Need to review this part ---

Oracle Database@AWS Network supports AWS's Shared VPC capability. Shared VPCs allow organizations to connect resources from multiple projects to a common Virtual Private Cloud (VPC) network in a Host project. Organization

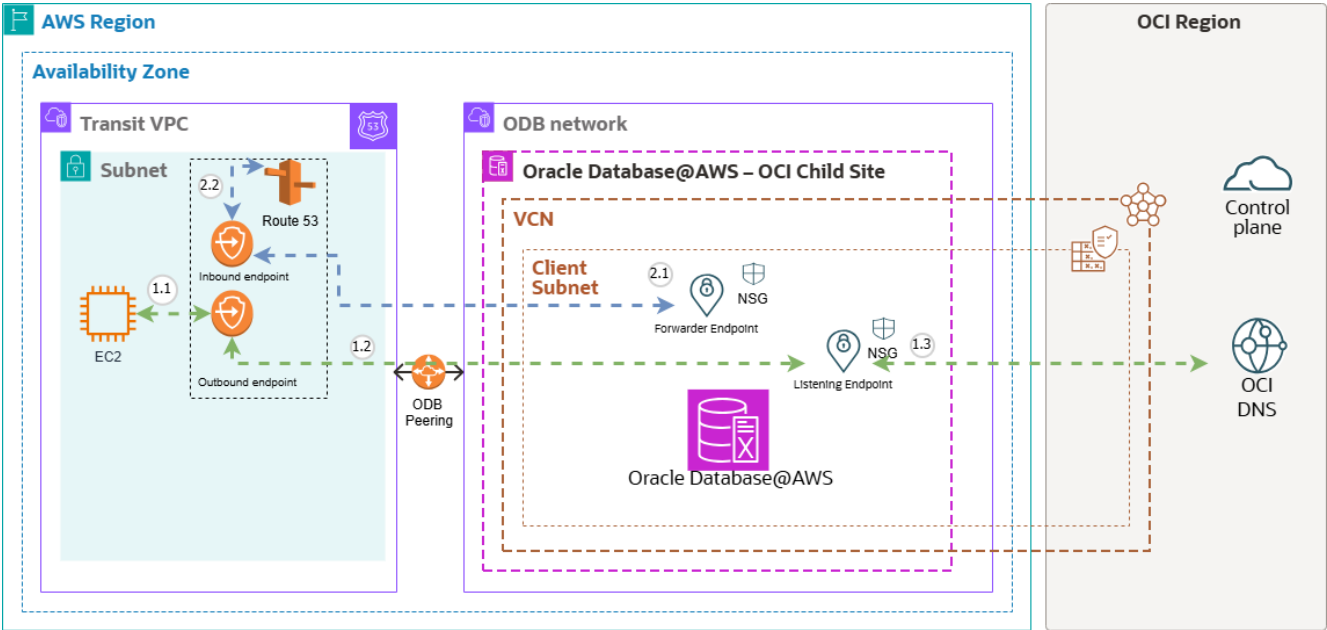
administrators can delegate management of applications to Service Project Administrators while maintaining centralized control over network resources. This simplifies the configuration and allows these resources to communicate securely and efficiently with each other using an internal IP address from the Shared VPC network. Service owners, like database users, use the network configuration defined by their network teams, instead of manually configuring the networks between different resources. It reduces the need to enable and configure VPC Peering and associated configurations.

Oracle Database@AWS Shared VPC architecture

Oracle Database@AWS Shared VPC architecture

DNS resolution

In addition to network topology for resource placement, describe any requirements with regards to DNS:



Oracle Database@AWS DNS

DNS resolution in AWS VPC is forwarded to OCI DNS Private Resolver. Configure DNS zone peering to resolve DNS entries in other VPCs.

3.5.4 Reliability and Resilience / Backup and Recovery

Please describe your solution from a backup and recovery design point of view. What should be the backup destination storage – OCI Object Storage or AWS Blob Storage or Any other target Automatic vs Manual backups. The Oracle-recommended backup solution is Zero Data Loss Recovery Service.

--- Need more detailed diagram --- Oracle Database@AWS Backup to OS Automatic Backup to OCI Object Storage

--- Need more detailed diagram --- Oracle Database@AWS Backup to Amazon S3 Automatic Backup to Amazon S3

--- Need more detailed diagram --- Oracle Database@AWS Backup to RCV Automatic Backup to Autonomous Recovery Service (located within OCI region or AWS region)

Backup Option	Info	Retention Period	Backup Destination
Oracle managed backup	Default and recommended		
From OCI cloud tooling	Automatic backup Max 60 days retention		

Backup Option	Info	Retention Period	Backup Destination
Standalone backup = until deleted	OCI Object storage (Oracle Managed) Charged from AWS credits		
User configured backup	Configure backups from the host using the dbaascli database backup and dbaascli database recover commands.	Defined by customer	OCI Object storage (Customer managed)
Charged from OCI credits			
Autonomous Recovery Service (recommended)	Recovery Service is the recommended option for backups with ExaDB-D	Retention period can be defined by customer. Long-term backups are available for up to 10 years	OCI region or AWS region
Manual setup	RMAN (Recovery Manager) or other backup tools can be configured to do backup to AWS Blob Storage or OCI Object storage	Defined by customer	To AWS Blob storage or OCI Object storage
Charged from respective CSP			

3.5.5 Monitoring

Guide:

Please describe your solution for monitoring options

- OCI Database Management Service
- OCI Ops Insights
- OCI Observability
- Oracle Enterprise Manager
- AWS CloudWatch
- AWS CloudTrail
- AWS EventBridge
- 3rd party tools (Splunk etc)

3.5.6 Governance

Guide:

Please describe your Oracle Database@AWS solutions' governance model

Oracle Database@AWS Governance Model

Oracle Database@AWS Governance Model

Oracle Database@AWS Identity Federation

Oracle Database@AWS Identity Federation

3.5.7 Manageability and Observability

The example below is the original OCI SDD Template text. For Oracle Database@AWS customers who want to leverage (parts of) OCI Observability and Monitoring, use the relevant sections of that text. For Oracle Database@AWS customers who want to include other forms of monitoring, describe those here if relevant.

Example:

Refer to the manageability-sol-con Snippet

3.5.8 Oracle Database@AWS Operating Model

Guide:

This section captures requirements on how the system will be managed after implementation and migration. Reference can be made to the normal “Explanation of Cloud Operations Service” for ExaDB-D in [MOS Note 2875973.1](#). The split between Oracle and customer responsibility is the same for ExaDB-D running within AWS as it is for ExaDB-D in OCI.

Example:

MYTestCustomer’s full AWS estate, including Oracle Database@AWS is managed fully by . For the management of the Exadata Database Service on Dedicated Infrastructure there is a split between Oracle responsibility and Customer/Partner responsibility, as described in “Explanation of Cloud Operations Service” for ExaDB-D in [MOS Note 2875973.1](#).

In short Oracle owns and is responsible for managing the infrastructure layer, and the customer/partner manages the layers on top of that (VMs, Databases, including access to these).

For an overview of responsibilities between AWS, Oracle and the end customer, see section “Roles and Responsibilities for Oracle Database@AWS” of the document [Exadata Database Service on Dedicated Infrastructure Security Controls](#).

For incident management, Oracle Cloud Support is the first line of support for all Oracle Database@AWS and questions. Detailed information on Oracle and AWS support responsibilities are available [here](#).

3.6 Sizing and Bill of Materials

Guide:

Estimate and size the physically needed resources of the solution. The information can be collected and is based upon previously gathered capacities, business user numbers, integration points, or translated existing on-premises resources. The sizing is possibly done with or even without a Physical Architecture. It is okay to make assumptions and to clearly state them!

Clarify with sales your assumptions and your sizing. Get your sales to finalize the BoM with discounts or other sales calculations. Review the final BoM and ensure the sales are using the correct product SKUs / Part Number.

Even if the BoM and sizing were done with the help of Excel between the different teams, ensure that this chapter includes or links to the final BoM as well.

Example:

3.6.1 Bill of Materials scope

This section covers only the Oracle Database@AWS elements, and any additional OCI elements. It does not contain any AWS services or components.

3.6.2 Collected Data for Sizing

Guide:

With OCCA Oracle Cloud Capacity Analytics the data collection comes from Oracle Enterprise Manager, AWR (Automatic Workload Repository) and/or is manually added.

Customer data is gathered with a script provided by Oracle with clear instructions on how to execute. It’s customer responsibility to run the extract script and optionally obfuscate the data and then upload the output to Oracle using secure methods.

The output needs a first validation, where any anomalies in the data collection, such as errors, missing data, unrealistic outliers/deviations or unexpected growth can be identified.

This data is then grouped in Cohorts which are in the context of sizing grouping of databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*
- *Workload isolation*
- *Application / User separation*
- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

Oracle Database@AWS Sizing Scope

Oracle Database@AWS Sizing Scope

Document the scope, manual additions, corrections and/or assumptions made related to the anomalies.

Example:

The scope that has been studied considers all the Oracle databases within the customer. This scope includes both Oracle Databases hosted on on-premise Exadata Infrastructures and Oracle Databases on legacy infrastructure (VMWare).

To complete the study, Oracle provided customer with scripts to capture the activity of the Oracle databases included in this scope. However, the activity of a number of databases could not be recovered. For these databases, we relied on assumptions shared by customer.

Oracle Database@AWS Sizing Graph

Oracle Database@AWS Sizing Graph

The data collection showed vCPU growth from August 2024 to September 2024 was higher than expected and because of that checked with and confirmed by customer. This will result in adjusted sizing values for OCPU.

3.6.3 Defined VM Clusters for Sizing

Guide:

Cohorts in the context of sizing are grouping databases together for the purpose of being deployed in VM Clusters. In the sizing process customer define Cohorts/VM Clusters for each database taking the following into consideration:

- *Network segmentation due to security*
- *Workload isolation*
- *Application / User separation*
- *Required Service levels (for example MAA Gold, Silver, Bronze)*
- *Size of the workload*
- *Physical Location*

Document the VM Clusters that has been agreed with the customer and any assumptions made. If you have a more complex VM Cluster layout with VM Cluster Node Subsetting, then consider visualizing it.

Example:

Based on the customers' different requirements for Production and non-Production environments which include different service levels as well as separate Client network segments for the environments, the following Cohorts (VM Clusters) have been defined.

Each VM cluster runs a single version of the Oracle Grid infrastructure (GI) to be chosen by the customer among the most recent. Each cluster can run different versions of Oracle Homes. Each independent VM cluster has OCPU assigned during the deploy phase of the VM cluster, so, we can assign different number of OCPU to different VM clusters and environments.

VM Clusters to be deployed are listed in the following tables together with the criteria that define why separate VM Clusters are deployed.

VM Clusters	Environment	VNet	MAA Service Level
Production A Primary	Production	DBVNetProd	Gold
Non-Production B	Test	DBVNetNonProd	Silver
Miscellaneous D	Test	DBVNetMisc	Silver

An ExaDB-D@OCI X11M Quarter Rack (with two DB Servers) can host up to X independent VM Clusters, the maximum number of VM Clusters is Y for X11M systems with more than 2 database servers. The maximum number of VMs per database server is Z regardless of the number of database servers in the system.

Documentation on in ExaDB-D@AWS is available [here](#)

3.6.4 Sizing Analysis

Guide:

Each VM Cluster is sized independently using Time-Aligned Peak Sizing with adjustments made for historical deviations and finally all VM Clusters are combined per ExaDB-D rack.

Values of interest include:

- OCPU Peak
- OCPU 95th Percentile
- DB Memory
- Allocated and Uses Storage
- IOPS Read and Write

Document the summary from the Sizing Summary for customer reference. Example: These are the time aligned plots for the two production VM Clusters of Database Service A (Primary + Standby).

Oracle Database@AWS Sizing Graph

Oracle Database@AWS Sizing Graph

Oracle Database@AWS Sizing Graph

Oracle Database@AWS Sizing Graph

3.6.5 Sizing Assumptions

Guide:

Document the customer requirements that is included in the sizing as well as assumptions made for the sizing. This information is provided as part of the Sizing Summary document.

Example:

The following customer requirements and general assumptions have been used in the sizing.

- All analyses are based on previous resource consumption and does not necessarily predict the future workload behavior.
- The defined cohorts (VM Clusters) and databases can differ from the current deployment.

- The sizing computations are based on the hourly average resource consumption of databases in each cohort.
- Cohort metrics are aligned by time and then the sum of peaks is computed for each hour.
- Hourly data is adjusted based on peaks found in the daily data over the past year (valid for EM data case).
- Using in-memory or other database options may reduce/increase the required processing power.
- Using different execution plans will influence the required processing power.
- All Prod DBs will have DR. 5% YoY growth was considered for a period of 4 years (corresponding system shape was selected).
- 10% additional overhead has been considered for converting standalone DBs to RAC.
- For each VM Cluster, an additional 4 OCPUs were added for OS+GI. Metrics for DR DBs were considered as 20% of the corresponding Primary DBs

3.6.6 Proposed Hardware Shapes

Guide:

Document the proposed hardware shape and any additions, corrections or customer requested changes made.

Example:

The sizing shows that ExaDB-D@AWS X11M Quarter Rack will fulfill requirements, but due to heavy use of CLOB in some Production Databases the customer requests to add an additional ExaDB-D@AWS X11M Storage Server to each system. This is referred to as an Elastic Configuration.

3.6.7 Licensing models and Elastic Expansion of Exadata on Oracle Database@AWS

Guide:

Fully elastic expansion of Exadata on Oracle Database@AWS, allows customers to provision the database compute and storage resources for any workload and provides the scale for any size database DB Systems.

Exadata on Oracle Database@AWS provides Multi-VM-Clustering and Node-Subsetting capabilities to provide workload segregation and resource isolation on a single Oracle Database@AWS infrastructure. The features are as follows:

- *Provision VM-cluster on a Subset of DB servers
- *Expand an Existing VM Cluster by Adding VMs
- *Shrink an Existing VM Cluster by Removing VMs

Oracle Database@AWS offers elastic infrastructure shapes to support workloads of different sizes. Customers can deploy flexible shapes that range from a Quarter Rack, 2 database and 3 storage servers, up to 32 database and 64 storage servers to meet a variety of CPU processing and storage requirements. Hence making it a perfect solution to consolidate on.

There are two licensing models:

1. Bring your own license (BYOL). Customer is only allowed to use the existing option licensed for.
2. License included. This includes all RDBMS options.

The licensing model is set at Virtual Cluster level and can be changes at any point during the duration of the subscription. This is very effective in scenarios where more licenses are required due to the scope growth of the project and/or customer has a desire not to maintain on-premises licenses.

Example:

This is the proposed Consumption Plan over five years based on the sizing, growth and other assumptions.

Oracle Database@AWS Sizing Estimate

Oracle Database@AWS Sizing Estimate

3.6.8 Exadata Database@AWS Bill of Materials

Product	Metric	Qty
B93380 - Exadata Cloud Infrastructure - Quarter Rack - X11M	N/A	2
B88592 - OCI - Database Exadata OCPU	OCPU BYOL	152
B91627 - Object Storage - Requests#[0-5]	10,000 Requests per Month (first 50,000 free)	1
B91628 - Object Storage - Storage#[10-999999999]	Gigabyte Storage Capacity Per Month	1000

Project Implementation (Only for Oracle Implementations!)

4.1 Solution Scope

NOTE

This section of the Oracle Database@AWS Exadata template is presented unaltered from the standard SDD Template (apart from this note).

For Oracle Implementations of an Oracle Database@AWS solution we would expect that the Oracle team only touches the specific Oracle/OCI elements of the solution. Therefore these can still be described from the OCI viewpoint, as per the standard template.

If the Oracle team also touch/implement non-OCI elements, then adjustments to this section will likely have to be made.

4.1.1 Disclaimer

Guide:

A scope disclaimer should limit scope changes and create awareness that a change of scope needs to be agreed upon by both parties.

Example:

4.1.2 Overview

Guide:

Describe the scope of the implementation as a sub-set of the solution scope. For example one environment from one application.

Example:

- Design and configure “least privilege” access controls and enable user access using OCI IAM compartments, groups, and policies.
- Design and provide a secure, scalable OCI network architecture.

4.1.3 Business Value

Guide:

Example:

4.1.4 Success Criteria

Guide:

*Technical success criteria for the implementation. As always be S.M.A.R.T: Specific, Measurable, Achievable, Relevant, Timebound. Example: 'Deployment of all OCI resources for the scoped environments in 3 months'.

The success criteria below are an example - Please replace them with your tailored success criteria.*

Example:

The below-listed success criteria are for the <Service Provider> implementation only. Partner activities and success criteria are not listed in this documentation.

- Finish provisioning of all OCI resources
- Establish all required network connectivity
- Successfully pass all test cases
- Finished handover with documentation
- Complete the Implementation Security Checklist

4.2 Workplan

4.2.1 Deliverables

Guide:

Describe deliverables within the implementation scope. Including this documentation as Solution Definition and the later following Solution Design. This should be a generic reusable text, provided by the implementers.

4.2.2 Included Activities

Guide:

Example: The implementation scope of work includes the following activities:

OCI Foundation & Network

- OCI Foundation Setup - 1 Region (REGION NAME)
- OCI Networking configuration
 - Creation of VCN for up to 3 environments (up to 12 VCNs total)
 - DRG and inter-VCN routing
 - Deployment of standard Security lists and NSG in VCN
 - Deployment of Route Tables in VCNs
- Configure one site-to-site IPSec VPN between OCI & on-premises
- Configure Web Application Firewall to route the incoming internet traffic to Load Balancers and configure recommended rules
- Configure bastion service to allow admin users to connect to the tenancy through the internet access

Security

- Enable Cloud Guard
- Enable Datasafe and Register the Databases in scope
- Enable VSS
- Configure OCI IAM Domains

Database

- Migrate one non-prod database with one iteration
- Migrate one prod database with two iterations

4.2.3 Recommended Activities

Guide:

All activities not stated in the [Included Activities](#) are out of scope, as described in the [Disclaimer](#). We do not provide a list of excluded activities to not create expectations based on a grey area between included and excluded activities. Here we only recommend further activities that happen to not be included but are not a full list of excluded activities.

Example:

All items not explicitly stated to be within the scope of the implementation project will be considered out of scope. Oracle recommends the use of professional services to implement extensions or customizations beyond the original scope, or to operate the solution with any of Oracle's certified partners. As a part of this engagement, the below activities are considered to be out of implementation scope.

- Any activities at customer on-premises or existing data center e.g. patching & backups required for migration
- Any integration with other products than in scope
- Any backup and recovery strategy implementation including third-party backup tool implementation
- Application upgrade of any Oracle or other vendor or open source software.
- SSL certificate management and configuration
- Any form of testing and validations, including but not limited to performance testing, load testing, HA testing, DR testing, and tuning of any component in the solution
- Any vulnerability assessment and penetration testing including server hardening, audit certification implementation
- Any functional testing is to be conducted by the customer and/or third party involved
- Any third-party firewall implementation, security tools, monitoring tools implementation
- Troubleshooting existing open issues, including the performance of the application
- Training on deployed products and OCI services
- Run and maintain the support of the environment and end-user training

4.2.4 Timeline

Guide:

4.2.4.1 Phase 1:

4.2.4.2 Phase n:

4.2.5 Implementation RACI

Guide:

Example:

Num	Activity	Oracle	Customer
1	Conduct Project Kickoff	AR	C
2	Provide access to the source environment, including all the relevant ports opened	I	AR
3	Provide VPN credentials for Oracle team, OCI console access details	I	AR
4	Prepare Source System, apply required patches on source environments for migration, and take source environment backup to OCI	I	AR
5	Backup of source Database	C	AR
6	Provision Landing Zone with related Network and policies in scope	AR	C
7	Configure site-to-site VPN between onPrem and OCI tenancy	AR	C

Num	Activity	Oracle	Customer
8	Migrate non-Prod database in scope	AR	C
9	Perform Pre and Post functional migration tasks	I	AR
10	Perform functional/customization/integration testing and Validation of application within the project timeline	I	AR
11	Provide OCI technical support during validation	AR	C
12	Prepare production runbook and perform Production Cutover	C	AR
13	Provide timely support for HW, OS, network related issues at source	I	AR
14	Procure of SSL Certificates	C	AR
15	Provide access to My Oracle Support required for product support along with CSI number	I	AR

R- Responsible, A- Accountable, C- Consulted, I- Informed

4.2.6 Assumptions

Guide:

List any assumptions, if any, which could impact the solution architecture or the implementation.

Example:

Generic assumptions

- It is assumed that all required contractual agreements between Oracle and the Customer are in place to ensure uninterrupted execution of the project.
- It is assumed that all work will be done remotely and within either central European time or India Standard Time normal office working hours.
- It is assumed that upgrades are excluded from the scope of work and no production systems/production cutover is part of the scope of work undertaken by the Oracle Service
- It is assumed that all required Oracle cloud technical resources are available for use during the duration of the project and that engineers involved have been granted the appropriate access to those technical resources by the customer before the start of the project.
- It is assumed that all required customer resources, and if applicable third-party resources, are available during the duration of the project to work openly and collaboratively to realize the project goals uninterruptedly.
- It is assumed that all required customer resources, and if applicable third-party resources are aware of all technical and non-technical details of the as-is and to-be components. All resources are committed to technical work as far as is needed for the execution of the project.
- It is assumed that all required documentation, system details, and access needed for the execution of the project can be given/granted to parties involved when and where deemed needed for the success of the project.
- It is assumed that the customer will have adequate licenses for all the products that may/will be used during the project and that appropriate support contracts for those products are in place where the customer will take the responsibility of managing any potential service request towards a support organization to seek resolution of a problem.
- It is assumed the customer will provide the appropriate level of information and guidance on rules and regulations which can directly and/or indirectly influence the project or the resulting deliverables. This includes, however not limited to, customer-specific naming conventions, security implementation requirements, internal SLA requirements as well as details for legal and regulatory compliance. It will be the responsibility of the customer to ensure that the solution will adhere to this.
- It is assumed that under the customer's responsibility, the customer will ensure and validate that the solution will be placed under the proper controls for ensuring business continuity, system availability, recoverability, security control, and monitoring and management as part of a post-project task.

- It is assumed that the customer will take responsibility for testing all functional and non-functional parts of the solution within the provided timeline and ensure a proper test report will be shared with the full team (including customer, Oracle, and if applicable third party).
- It is assumed that any requirement, deliverable, or expectation that is not clearly defined as in-scope of the project will not be handled as part of the project and is placed under the responsibility of the customer to be handled outside of the project.

Project-specific assumptions

- It is assumed that sufficient network bandwidth (greater than 200 GB) is available between OCI and Customer onPremise for any data transfer.
- It is assumed that the customer, or a partner of your choice, will own the control, access, management, and further development of your OCI environment following the deployment of your solution.

4.2.7 Obligations

Guide:

Example:

- You will have purchased the appropriate Universal Credits for the services required for the project.
- The implementation team will have admin access to the customer's tenancy for implementation.
- You will ensure the appropriate product training has been obtained to maintain and support the implementation
- Your business team will be available for the Testing phase, which will be completed within the agreed testing window.
- You will provide project management for the project and will manage any third-party suppliers or vendors.
- You will provide the implementation team with appropriate access to your tenancy & relevant on-premises applications/database to perform implementation activities. We recommend the least-privilege access principle.
- You will revoke implementor access on production goLive or after project completion.
- You will take consistent and restorable backups of your existing data and application before implementation.

Add for EBS migration

- Your on-premise source non-prod environment would be a fresh clone from the production environment for easy simulation of issues.
- You would be responsible for applying and testing all migration-related patches on the on-premise source environment.
- You will ensure that the relevant pre-requisite patches have been applied on the on-premise source environment as per MOS DocID 2517025.1: Getting Started with Oracle E-Business Suite on Oracle Cloud Infrastructure:
 - Table 5 - Source Environment Requirements with Target Database Tier on Oracle Cloud Infrastructure Compute VM (Under 4.2.2 section) and
 - Section 4.5.5 Applying the Latest Critical Patch Updates (CPU) and Security Fixes

4.2.8 Transition Plan

Guide:

The Transition Plan describes the handover of the project, after the implementation. Please ensure the accepting transition party is filled out.

Annex

5.1 Security Guidelines

5.2 Networking Requirement Considerations

5.3 Networking Solutions

5.4 Manageability

5.5 Onboarding